LAB 4: TokTickIT Service Desk Workflow and Dashboards

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Report edition: Evidence edition, 1 November 2026. Lab 4 Issues #51–#56 / PRs #57–#62, UX/UI Issue #63 / PR #64, and responsive follow-up Issue #65 / PR #66 are closed and merged. The latest merge is `a76ee09dc558aa2efcd6f4ef3f3bd759ba3bed7e`, verified as the current main tip. This report records evidence by the source revision it covers and does not claim a guaranteed mark.

Answer Part 1: Git Use with Engineering Workflow (10 Points)

1.1 Repository, Project, Issue and PR URLs

Evidence	Verified destination
GitHub repository	HolyThiccDaddy/toktickit
GitHub Project board	TokTickIT project #1
Lab 4 Issues	#51 · #52 · #53 · #54 · #55 · #56
Lab 4 final integration	PR #62 → main
UI rework follow-up	Issue #63 / PR #64
Tablet responsive correction	Issue #65 / PR #66
Lab 4 source documents	docs/lab-04 on main
README / .gitignore	README · .gitignore

1.2 Ordered feature branches and peer approvals

Issue	Feature branch → base	PR	Merged SHA (short)	Peer review
#51 1: Engineering contract and specification/test DD	feature/lab4-issue-1-contract → lab4-staging	#57	083b1a4367	Approved by osizk; merged by osizk
#52 2: Actions Taken database, migration, seed, and API	feature/lab4-issue-2-actions → lab4-staging	#58	2e4ddf645d	Approved by osizk; merged by osizk
#53 3: Actions Taken UI	feature/lab4-issue-3-actions-ui → lab4-staging	#59	0253c668d2	Approved by osizk; merged by osizk
#54 4: Ticket workflow and resolution gate	feature/lab4-issue-4-ticket-workflow → lab4-staging	#60	1a3591421d	Approved by osizk; merged by osizk
#55 5: Requester, IT Staff, and Administrator dashboards	feature/lab4-issue-5-dashboards → lab4-staging	#61	026ba36348	Approved by osizk; merged by osizk
#56 6: Final regression, accessibility, visual evidence, release PDF, and integration	feature/lab4-issue-6-final-release → main	#62	96a34538ef	Approved by osizk; merged by osizk

PRs #57–#61 integrate the ordered feature slices into lab4-staging; PR #62 integrated the Lab 4 release into main at `96a34538efcb6675ac8129fd7e7b7c63e619e724`. The corresponding Issues #51–#56 are closed. The UI rework PR #64 and responsive follow-up PR #66 were separately merged to main; GitHub's main branch now points to `a76ee09dc558aa2efcd6f4ef3f3bd759ba3bed7e`. The Project board could not be independently re-read from the connected account, so this report does not infer its card columns from closed Issues.

1.3 Follow-up branch records

Issue	Branch → base	PR	Head SHA	Merge SHA	Review
#63 · UX/UI	codex/63-ux-ui-rework → main	#64	ca82049e1681	e22a50e3a932	Approved / merged
#65 · Tablet overlap	codex/65-tablet-ticket-operations → main	#66	097cdca6ba6e	a76ee09dc558	Approved / merged

1.4 UI rework history and responsive follow-up

Issue #63 / PR #64 addressed the UX/UI rework and restored permitted direct URLs; it was approved by @osizk and merged at `e22a50e3a93250567e9722e3ce4c4c6402458c81`. Subsequent final responsive QA identified overlapping Ticket-operation controls at tablet widths. Issue #65 / PR #66 changed the Staff Ticket Detail grid and added regression assertions at 768px, 991px and 992px. PR #66 was formally approved by @osizk and merged to main at `a76ee09dc558aa2efcd6f4ef3f3bd759ba3bed7e`; Issue #65 is closed. Part 3 separates the server run on the preceding main tree from the client and full E2E results recorded for the PR #66 follow-up.

1.5 Main branch and review evidence

The verified main branch tip is `a76ee09dc558aa2efcd6f4ef3f3bd759ba3bed7e`, the PR #66 merge commit. The earlier PR #64 main SHA `e22a50e3a93250567e9722e3ce4c4c6402458c81` remains documented as a distinct release milestone. GitHub returned zero status checks and zero workflow runs for the final merge SHA; local tests in Part 3 are not presented as GitHub CI.

1.6 Rendered peer-review evidence

PR / linked Issue	Peer feedback and author re-review	Formal decision
PR #57 / Issue #51	Action lifecycle, resolution evidence, dashboard count/drill-down parity, and complete AC mapping. Peer review · Author re-review	APPROVED Merge reply
PR #58 / Issue #52	Seed rerun preservation and idempotent replay after assignee deactivation. Peer review · Author re-review	APPROVED Merge reply
PR #59 / Issue #53	Follow-up value clearing, timestamp precision, revision history, and eligible assignees. Peer review · Author re-review	APPROVED Merge reply
PR #60 / Issue #54	Reject unsupported stale preconditions and recover Action evidence after load failure. Peer review · Author re-review	APPROVED Merge reply
PR #61 / Issue #55	Route My Actions correctly and match dashboard predicates/date windows. Peer review · Author re-review	APPROVED Merge reply
PR #62 / Issue #56	Nine-part PDF criteria, awaited metric data, and exact SHA-bound command output. Peer review · Author re-review	APPROVED Merge reply
PR #64 / Issue #63	Restore permitted direct URLs and retain queue/date filters in new-tab navigation. Peer review · Author response recorded	APPROVED Merge reply
PR #66 / Issue #65	Fix tablet overlap between assignment and IT Priority controls; add boundary-width regression assertions. Peer review · Author response recorded	APPROVED Merge reply

The links show peer findings, author re-review responses, formal approvals, and merge acknowledgments for the six ordered Lab 4 PRs plus the two separate UI follow-ups. The full review history is in reviewer.md. Issue closure and PR merge are verified; Project-card Done remains a separate unverified gate.

1.7 Repository structure and delivery hygiene

```
toktickit/ ■■■ client/ · React UI, component tests, and Playwright dependency ■■■ server/ · Express API, Prisma schema/migrations, and API tests ■■■ e2e/ · Playwright journeys and deterministic setup/teardown ■■■ docs/lab-02/ · prior-lab contract and test evidence ■■■ docs/lab-03/ · authenticated workflow contract and evidence ■■■ docs/lab-04/ · specification, API/UI contracts, tests, AI-use and reviewer record ■■■ artifacts/lab-04/ · verification output and real screenshots ■■■ output/pdf/ · final submission report
```

README records the React/TypeScript, Express/Prisma, PostgreSQL stack and inherited Lab 1–3 behaviors. The committed .gitignore excludes dependency folders, local environment files/secrets, build output, local databases, logs, uploads, and Playwright outputs.

README

.gitignore

Answer Part 2: Specification Driven Development (Spec DD) (5 Points)

2.1 Sprint goal and explicit boundaries

Extend the authenticated service desk with traceable Actions Taken, a complete Ticket workflow and resolution gate, and role-scoped dashboards, while preserving Lab 1–3 behavior. The contract explicitly excludes SLA/escalation engines, external notifications, inventory/purchasing/cost accounting, multi-level approval/signatures, advanced BI/export warehouses, and unapproved production/multi-tenant features.

Full numbered specification.md

2.2 Functional requirements

ID	Requirement
FR-01–04	One Ticket to many chronologically listed Actions; authenticated Staff/Admin create and versioned edit; Requester reads only owned Ticket Actions.
FR-05–07	Respect public/internal visibility; conditional follow-up note; descriptive attachment notes reuse existing file-upload policy.
FR-08–10	Support all eight Ticket statuses; enforce role-specific transitions, terminal protection, stale-write handling, and resolution evidence server-side.
FR-11–14	Role-scoped Requester/Staff/Admin dashboards; concise server-side metrics and exact drill-downs; zero results remain explicit.
FR-15–16	Every protected API uses authenticated session, role/ownership checks, and safe conflicts for stale writes.
FR-17–20	Forward-compatible migration preserves prior Lab data; deterministic idempotent seed; Lab 1–3 regressions remain covered; all loading, empty, validation, success, forbidden, not-found, conflict and retry states are testable.

2.3 Action fields and lifecycle

Field	Contract
Action date/time	Required; timezone-aware; display in the documented application timezone.
Description / result	Trimmed; required; description 10–2,000 characters; result 1–2,000; whitespace-only rejected.
Performer	Derived from the authenticated server session; client actor IDs are ignored.
Assignee	Optional active IT Staff/Admin only; assignment does not change performer or audit history.
State / audit	OPEN → IN_PROGRESS / COMPLETED / CANCELLED; IN_PROGRESS → COMPLETED / CANCELLED; terminal states do not transition.
Follow-up	If required, note is required. Resolution records time and authenticated resolver, appends revision, increments version; Action remains COMPLETED.
Attachment notes	Optional descriptive text; does not upload a file or expose a storage key.
Edit history	Each accepted versioned edit stores a prior immutable snapshot; stale/rejected edits append no revision.

2.4 Ticket status transitions and resolution gate

From	To	Role	Confirmation / gate
NEW	OPEN	Staff/Admin	None
OPEN	IN_PROGRESS	Staff/Admin	None
IN_PROGRESS	WAITING_FOR_REQUESTER	Staff/Admin	None
WAITING_FOR_REQUESTER	IN_PROGRESS	Staff/Admin	None
IN_PROGRESS	RESOLVED	Staff/Admin	Confirm + qualifying completed Action + no outstanding follow-up
RESOLVED	CLOSED	Staff/Admin	Confirm + same resolution gate
RESOLVED	REOPENED	Staff/Admin	Confirm
REOPENED	IN_PROGRESS	Staff/Admin	None
NEW / OPEN / IN_PROGRESS / WAITING_FOR_REQUESTER / REOPENED	CANCELLED	Staff/Admin	Confirm; terminal afterwards

A qualifying Action must be COMPLETED, belong to the Ticket, contain non-empty description/result, and have completion time and actor. Every follow-up must also be resolved before RESOLVED/CLOSED. Requester status indications are advisory only. Any unlisted, unauthorized, stale, or terminal transition is rejected without mutation.

2.5 Dashboard metric predicates

Role / metric	Exact server predicate	Drill-down
Requester · Open	Owned by session; NEW, OPEN, IN_PROGRESS, WAITING_FOR_REQUESTER, REOPENED	/my-tickets?currentStatuses=…
Requester · Waiting	Owned by session; WAITING_FOR_REQUESTER	/my-tickets?currentStatus=…
Requester · Updated	Owned by session; now−7d ≤ updatedAt < now	/my-tickets?updatedFrom…&updatedTo…
Requester · Resolved	Owned by session; RESOLVED/CLOSED; last 7 UTC days	/my-tickets?currentStatuses…
Staff/Admin · Unassigned	ownerId IS NULL; active work statuses	/staff/queue?assigneeId=unassigned
Staff/Admin · My owned	ownerId=session; active work statuses	/staff/queue?assigneeId=me
Staff/Admin · In progress	IN_PROGRESS or WAITING_FOR_REQUESTER	/staff/queue?statuses=…
Staff/Admin · Updated	Accessible Ticket; now−7d ≤ updatedAt < now	/staff/queue?updatedFrom…
Staff/Admin · My Actions	performedBy=session; now−30d ≤ actionDateTime < now	/staff/actions?performedBy=me

2.6 Authorization matrix

Capability	Requester	IT Staff	Administrator
View Dashboard	Own metrics only	Operational metrics	Staff metrics + permitted account summary
Read Ticket	Own Tickets	Authorized queue/detail	Authorized queue/detail
Read Actions	Own Ticket history	Accessible Ticket history	Accessible Ticket history
Create / edit Action	Denied	Active session + CSRF + version	Active session + CSRF + version
Change Ticket status	Denied; response advisory	Allowed transition + server gate	Allowed transition + server gate
Administer users	Denied	Denied	Admin role + last-admin safeguard

2.7 Shared UI, accessibility and responsive contract

All roles reuse the Lab 2/3 Zen Green tokens and shared controls. Active navigation and keyboard focus remain visible. Inputs have accessible labels; status and priority include text cues rather than color alone; field errors connect to controls and announce through a live region. Recoverable failures preserve entered form values.

Desktop begins at 992px, tablet at 768–991px, and mobile below 768px. Tables and dashboard columns reflow within their component; the page itself must not scroll horizontally. Keyboard focus follows visible navigation, dialogs restore focus to their trigger, and reduced-motion preferences suppress decorative transitions.

API filters, auth matrix and dashboard contract

Responsive screen, component and state contract

2.8 Migration, definition of done, and spec-first evidence

Release contract	Acceptance rule
Migration	Forward-only, additive schema changes preserve Lab 1–3 records; seeded data is deterministic and rerunnable.
Verification	Protected APIs enforce session, role and ticket ownership; stale mutations conflict without partial writes.
Definition of done	Every numbered AC maps to an executable test path; full inherited regressions, responsive/accessibility evidence, recovery checks, docs and nine-part PDF are reviewed before final integration.
Out of scope	No SLA/escalation engine, external notification system, inventory/cost accounting, multi-level approval, or unapproved production feature.

Evidence	Verified timestamp / relationship
Initial spec + Test DD commit	8b8577781957d6079fcfbed421a77e2a5f6345c9 · 29 Oct 2026, 19:06 +07
Reviewed contract commit	2cf44f91c3fc949f5ce4b515e4218b60589b00ef · merged in PR #57 before implementation work
First implementation PR opened	PR #58 opened 30 Oct 2026, 20:59 UTC (31 Oct, 03:59 +07), after the reviewed contract was merged

PR #57 contract history

PR #58 implementation timeline

Answer Part 3: Test Driven Development and Traceability (Test DD) (10 Points)

3.1 Acceptance-criterion traceability

Test ID	AC	Coverage and executable file	Status
DOC-01	AC-01	Contract documents exist, cross-link, and contain the agreed Action/workflow/dashboard rules File: contract-docs.test.ts	PASS
API-01	AC-02, AC-03, AC-20	Create valid Action and enforce exact trimmed limits, whitespace, and conditional fields File: actions-taken.api.test.ts	PASS
API-02	AC-04	Requester read isolation and write denial File: actions-taken.api.test.ts	PASS
API-03	AC-05, AC-18	Current-version edit appends an immutable before-image; stale/rejected edit adds no revision and… File: actions-taken.api.test.ts	PASS
API-04	AC-06	Idempotent seed and zero/one/many Action fixtures File: actions-seed.test.ts	PASS
API-05	AC-07, AC-09	Complete status transition matrix and terminal protection File: ticket-workflow.api.test.ts	PASS
API-06	AC-08, AC-21	Resolution gate blocks on outstanding follow-up, then permits Ticket resolution after authorized… File: ticket-workflow.api.test.ts	PASS
API-07	AC-10	Requester metrics, UTC boundaries, zero/one/many, and equality between each dashboard count and the… File: requester-dashboard.api.test.ts	PASS
API-08	AC-11	Staff/Admin metrics, exact queue drill-down equality using assigneeId=me/unassigned, multi-status and… File: staff-dashboard.api.test.ts	PASS
API-09	AC-15	Existing records and Lab 1–3 behavior survive migration File: migration-regression.api.test.ts	PASS
UI-01	AC-12, AC-14, AC-18–AC-21	Actions list/form exact validation, assignment/state controls, follow-up resolution, immutable… File: ActionsTaken.test.tsx	PASS

Acceptance-criterion traceability · continued

Test ID	AC	Coverage and executable file	Status
UI-02	AC-10, AC-12	Requester dashboard metrics, ownership states, and exact Waiting/Open/Recently Updated/Recently… File: RequesterDashboard.test.tsx	PASS
UI-03	AC-11, AC-12	Staff/Admin dashboard metric-to-queue predicate equality, supported owner/unassigned/status/date… File: StaffDashboard.test.tsx	PASS
UI-04	AC-08, AC-09, AC-12	Status controls and resolution gate feedback File: TicketWorkflow.test.tsx	PASS
UI-05	AC-13, AC-14	Desktop/tablet/mobile layout and keyboard accessibility File: responsive.visual.spec.ts, release-evidence.visual.spec.ts	PASS
UI-STYLE-01	AC-12, AC-13, AC-14	Shared Zen Green tokens, accessible names, focus ring, and no-overflow style contract File: StaffDashboard.test.tsx, RequesterDashboard.test.tsx, release-evidence.visual.spec.ts	PASS
UNIT-01	AC-07–AC-09	Action state transitions and resolution-gate predicate as pure domain rules File: ticket-workflow.api.test.ts, TicketWorkflow.test.tsx	PASS
PERF-01	AC-10–AC-13	Dashboard queries and bounded drill-down remain within the documented smoke budget File: dashboard-performance-smoke.spec.ts	PASS
E2E-01	AC-02–AC-06, AC-17, AC-18, AC-20	Staff creates Actions with retry-safe keys, edits, retries, and Requester reads the prior-value timeline File: actions-taken-flow.spec.ts	PASS
E2E-02	AC-07–AC-09	Status workflow, resolution gate, reopen/cancel File: ticket-resolution.spec.ts	PASS
E2E-03	AC-10–AC-13	Requester/staff/admin dashboards and responsive widths File: dashboards.spec.ts, release-evidence.visual.spec.ts	PASS
A11Y-01	AC-12–AC-14	Labels, focus, status cues, live regions, no overflow File: responsive.visual.spec.ts, release-evidence.visual.spec.ts	PASS
EVID-01	AC-16	Final evidence records exact SHA, command output, screenshots, links, and the single nine-part PDF File: release-evidence.visual.spec.ts, LAB_04_FINAL_SUBMISSION_REPORT.md	PASS
API-10	AC-17	Missing/malformed key rejection, same-key/same-payload replay without duplicate row, and… File: actions-taken.api.test.ts	PASS
API-11	AC-19	Assign/reassign, full Action transition matrix, inactive target rejection, terminal protection, and… File: actions-taken.api.test.ts	PASS
E2E-04	AC-19	Staff assigns/reassigns an Action, completes or cancels through permitted transitions, and receives… File: actions-taken.api.test.ts, ActionsTaken.test.tsx	PASS
E2E-05	AC-08, AC-21	Ticket resolve is blocked by a completed Action's outstanding follow-up; authorized Staff resolves… File: ticket-workflow.api.test.ts, ActionsTaken.test.tsx	PASS

3.2 Executed Lab 4 test files

Suite	Executable test file	Passed / failed
Server	lab-04/actions-seed.test.ts	1 / 0
Server	lab-04/actions-taken.api.test.ts	30 / 0
Server	lab-04/contract-docs.test.ts	1 / 0
Server	lab-04/requester-dashboard.api.test.ts	2 / 0
Server	lab-04/staff-dashboard.api.test.ts	4 / 0
Server	lab-04/ticket-workflow.api.test.ts	6 / 0
Client	lab-04/ActionsTaken.test.tsx	7 / 0
Client	lab-04/AppDirectNavigation.test.tsx	3 / 0
Client	lab-04/RequesterDashboard.test.tsx	2 / 0
Client	lab-04/StaffActionsList.test.tsx	1 / 0
Client	lab-04/StaffDashboard.test.tsx	2 / 0
Client	lab-04/TicketWorkflow.test.tsx	4 / 0
Client	lab-04/WorkspaceShell.test.tsx	3 / 0

3.3 Verified suite outputs and commit boundaries

Suite / build	Observed result	Evidence binding
Server Vitest	44 files; 136 passed; 0 failed	Fresh run on ca82049e1681; server sources unchanged by PR #66 through main a76ee09dc558
Client Vitest	46 files; 73 passed; 0 failed	PR #66 verification; merged head 097cdca6ba6e; raw JSON retained
Playwright full E2E	36 passed; 0 failed; 0 skipped	Desktop, tablet and mobile; includes the 768/991/992px overlap assertions
Production builds	Server TypeScript build freshly passed; client TypeScript + Vite build passed for PR #66	Server verified after merge; PR #66 changed only client UI and E2E test
Responsive captures	27 Lab 4 release PNGs (9 screens × 3 viewports) plus 28 compliance captures	Follow-up evidence includes 7 explicit width-boundary captures
Migrations / seed	6 migrations; no pending migration; guarded _test seed twice	Operational-table fingerprints unchanged
GitHub checks	0 commit statuses; 0 workflow runs for a76ee09dc558	No GitHub CI success is claimed

Server tests and TypeScript build were freshly rerun from source commit ca82049e1681 after PR #66 had merged. PR #66 changed only `client/src/StaffTicketDetail.tsx` and `e2e/lab-04/compliance-responsive.visual.spec.ts`, so the server sources are unchanged through final main a76ee09dc558. Server: 44 files / 136 passed / 0 failed; build command `npm run build` passed. PR #66's verification record reports Client 46 files / 73 passed, the production build passed, and Playwright 36 passed / 0 failed / 0 skipped across desktop/tablet/mobile. Raw server output and run metadata: `artifacts/lab-04/verification/ui-rework/final-server-results.json` and `final-server-run.json`. GitHub returned no status checks or workflow runs for the final merge SHA; no CI pass is claimed.

PR #66 review, verification record and merge

3.4 Guarded migration, repeat seed and recovery rehearsal

The seed rehearsal loaded only server/.env.test, parsed the PostgreSQL URL, and verified the database name ends in `_test` before using Prisma. Six migrations were already applied. Two compiled seed runs completed; 13 operational tables were fingerprinted. The repeat-seed check excludes only inherited upsert timestamps `User.updatedAt` and `RequesterUser.updatedAt`; it does not claim byte-identical values for those two fields.

A separate report-time recovery rehearsal used only a new scratch schema whose name ended in _test, ran the six migrations and deterministic seed, dumped/restored the schema, verified all 13 table fingerprints and 10 sequences plus the seeded protected attachment bytes, then removed only its own scratch schema/temp folder. This rehearsal is supplemental and was not counted as a test suite or as production/main evidence.

The database examples are intentionally labeled with their sample counts. Seeded revision, public-comment and internal-note tables are empty in this fixture; the recovery rehearsal verified their table fingerprints, but this run does not assert non-empty revision/comment content.

3.5 Scratch recovery row-count snapshot

Table	Rows	Table	Rows	Table	Rows
ActionRevision	0	ActionTaken	8	Attachment	1
AuthSession	0	Category	4	InternalNote	0
PublicComment	0	RelatedSystem	7	RequesterUser	5
Ticket	8	TicketCounter	1	User	10
_prisma_migrations	6				

Answer Part 4: AI Use with Reflection (5 Points)

The contract record names OpenAI Codex (GPT-5) for the original Lab 4 work. The separate Issue #63 UI-conformance pass used OpenAI Codex (GPT-6); this does not retrospectively change the original model record.

#	Prompt focus	Decision / verification
1	Extract assignment scope, workflow, repository and PDF rules	Handout sections 3, 11–14; specification
2	Design Action Taken data model and audit	Authenticated performer; immutable revisions
3	Enumerate status matrix and resolution gate	Specification BR-11/12; API tests
4	Define dashboard metrics and drill-down equality	UTC bounds; API/UI equality tests
5	Review API authorization, CSRF, stale write and idempotency	Lab 3 session contract; server tests
6	Map acceptance criteria to test types	tests.md traceability table
7	Review desktop/tablet/mobile and accessibility	ui-spec and Playwright evidence
8	Audit release evidence against exact revision	SHA binding; no claim from unrun tests
9	Resolve peer comments on revisions, retries and limits	PR #57 review history; Lab 2/3 conventions
10	Resolve follow-up before Ticket resolution	BR-12a and API/UI/E2E workflow

4.1 My Reflection

I set the product direction and acceptance expectations. I asked Codex to compare the Lab 4 requirements with earlier labs, implement the planned slices, run checks, prepare screenshots and the report, and follow the branch, Issue-linking, and peer-review flow. For the UI, I supplied reference images and a video and specified the palette, typography, background, selected-state color, full-page layout, rounded corners, border contrast, responsive behavior, and transitions. I also pointed out when evidence or GitHub workflow details did not match what I had asked for, including the need to link each PR to its Issue.

Codex wrote and changed code and tests, ran commands, captured browser evidence, and prepared the report and pull requests; I did not personally perform every implementation or verification step. My contribution was setting the requirements, choosing the visual direction, reviewing the presented results, and asking for corrections when they missed the assignment or our agreed workflow. This process taught me to check the actual screens and the exact branch and commit behind a report: a green test count or a merged PR by itself does not prove every responsive state or submission artifact is correct. Before submitting, I need the final report and screenshots to match the reviewed code on main, and the Kanban state to match the actual review and merge history.

This reflection is based on the student's recorded directions in this conversation and should be checked for personal accuracy before final submission.

Full selected prompt record and reflection

Answer Part 5: Working IT Staff Dashboard UI (5 Points)

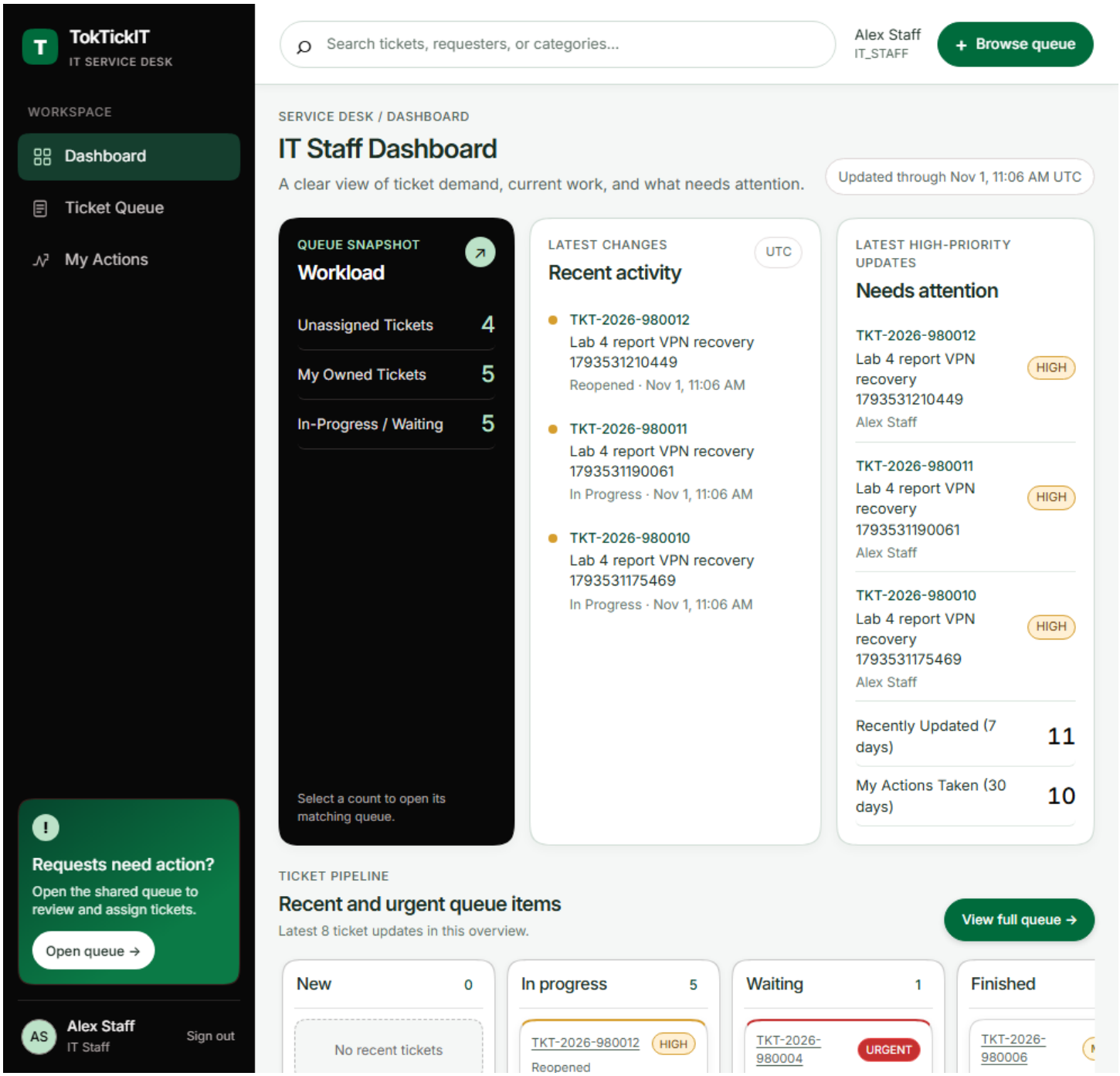
Requester metrics remain ownership-scoped. IT Staff and Administrator dashboards share an authorized operational view, including unassigned/currently-owned work, active queue counts, recent/urgent tickets, and My Actions. Cards link to filters with matching predicates; loading, empty, error/retry, forbidden and responsive states are represented in the suite and evidence.

5.1 Independent API and database comparison

Metric	Staff API	Independent Prisma count	Drill-down
Unassigned Tickets	4	4	Queue: unassigned + active statuses
My Owned Tickets	5	5	Queue: me + active statuses
In-Progress / Waiting	5	5	Queue: IN_PROGRESS / WAITING

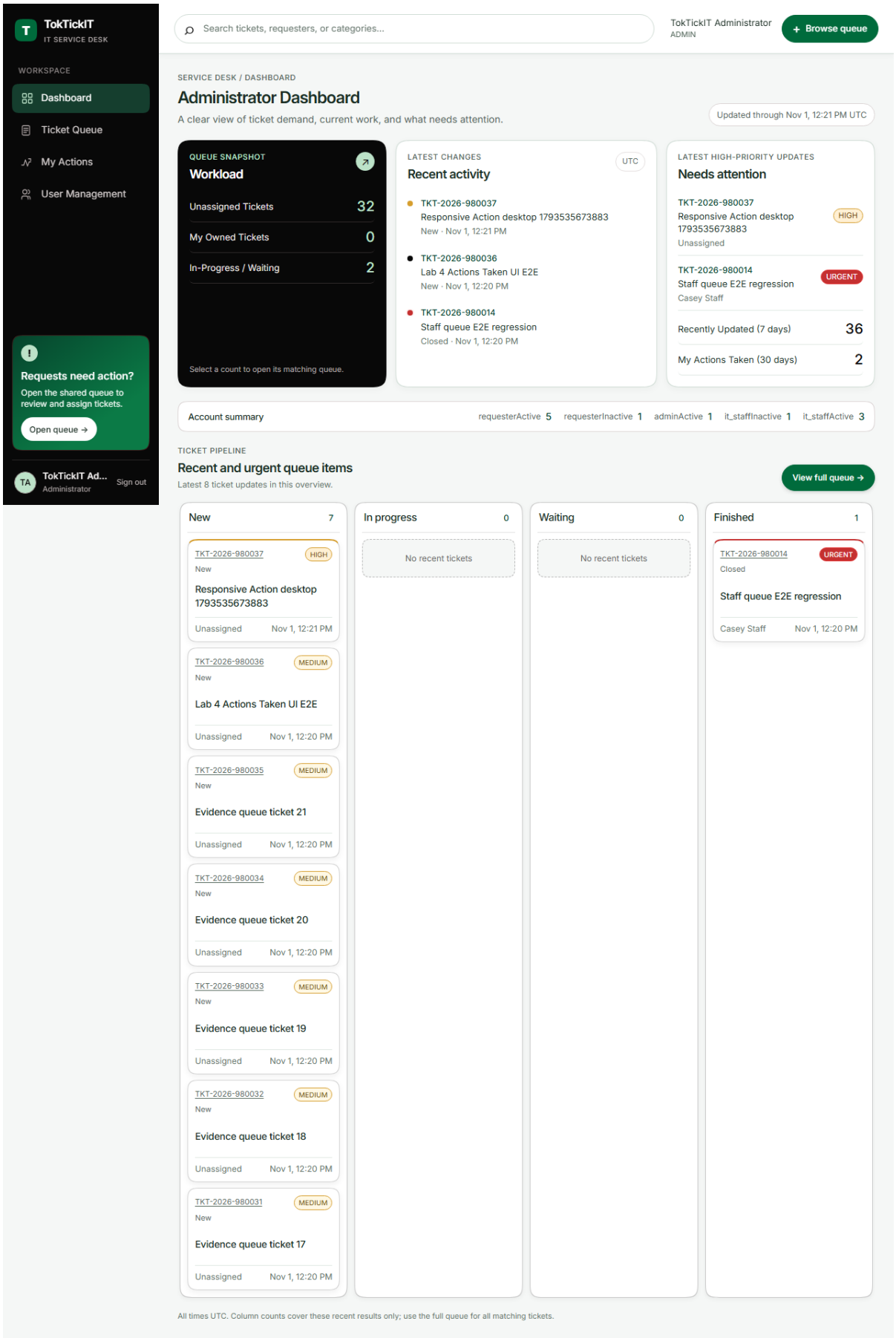
The counts were read as IT Staff, then independently recomputed from the guarded test database. These numbers are a fixture snapshot, not live production data; each API count equals its matching database count in this run.

IT Staff · report-time capture



1 Nov 2026 browser capture; shown metrics checked against independent Prisma counts.

Administrator · post-merge E2E



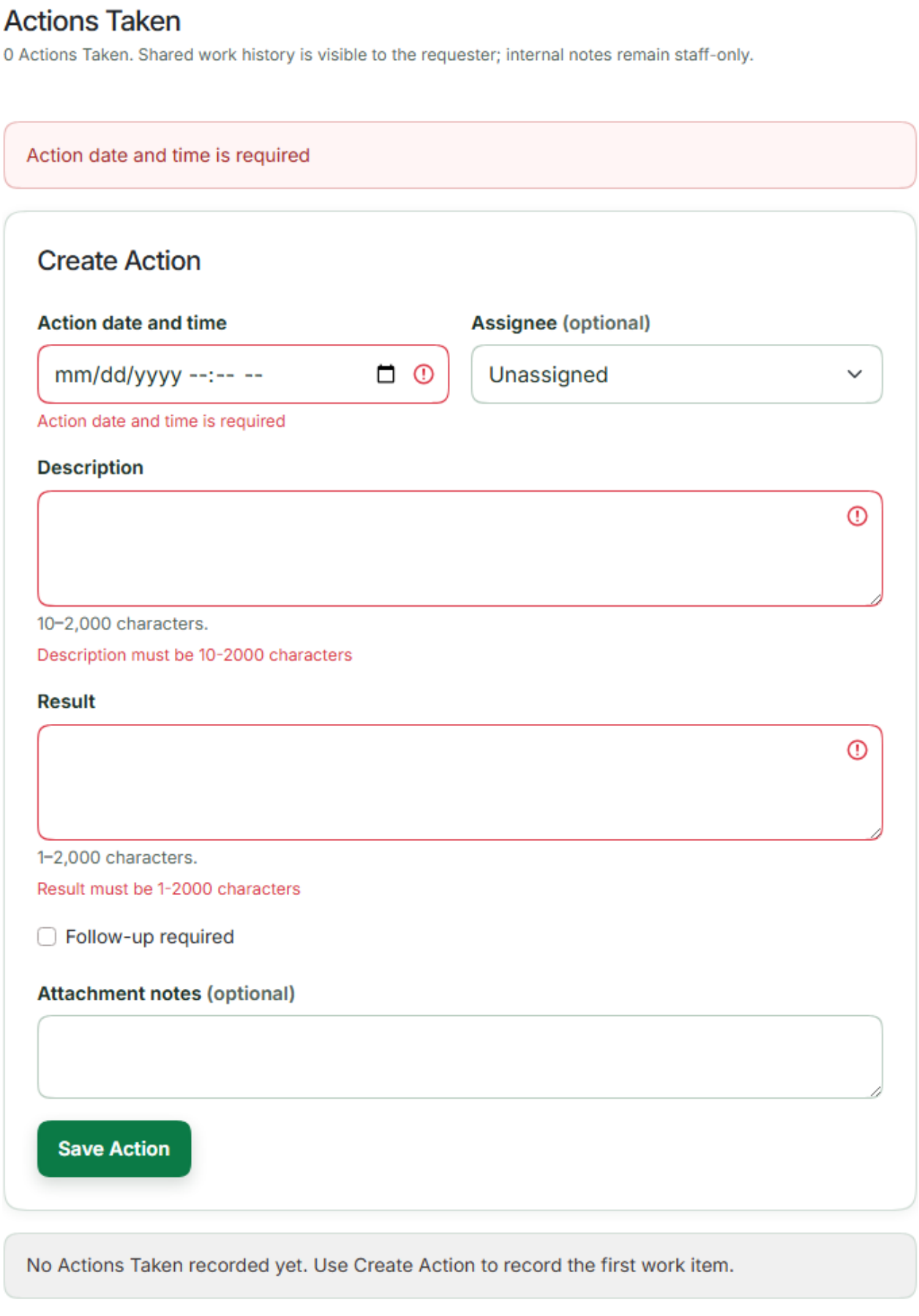
Fresh authenticated desktop capture after PR #64 merged to main.

Answer Part 6: Working Actions Taken UI (10 Points)

Authorized Staff/Admin users can create, assign/reassign, edit, complete and cancel Actions Taken. The form validates required values, preserves safe retry states, rejects inactive assignees, and shows immutable revision history. Requesters see a read-only history for their own Ticket. This report combines focused report-time browser evidence with the 36-test post-merge browser run; the full screenshot set is kept separately in the post-merge evidence folder.

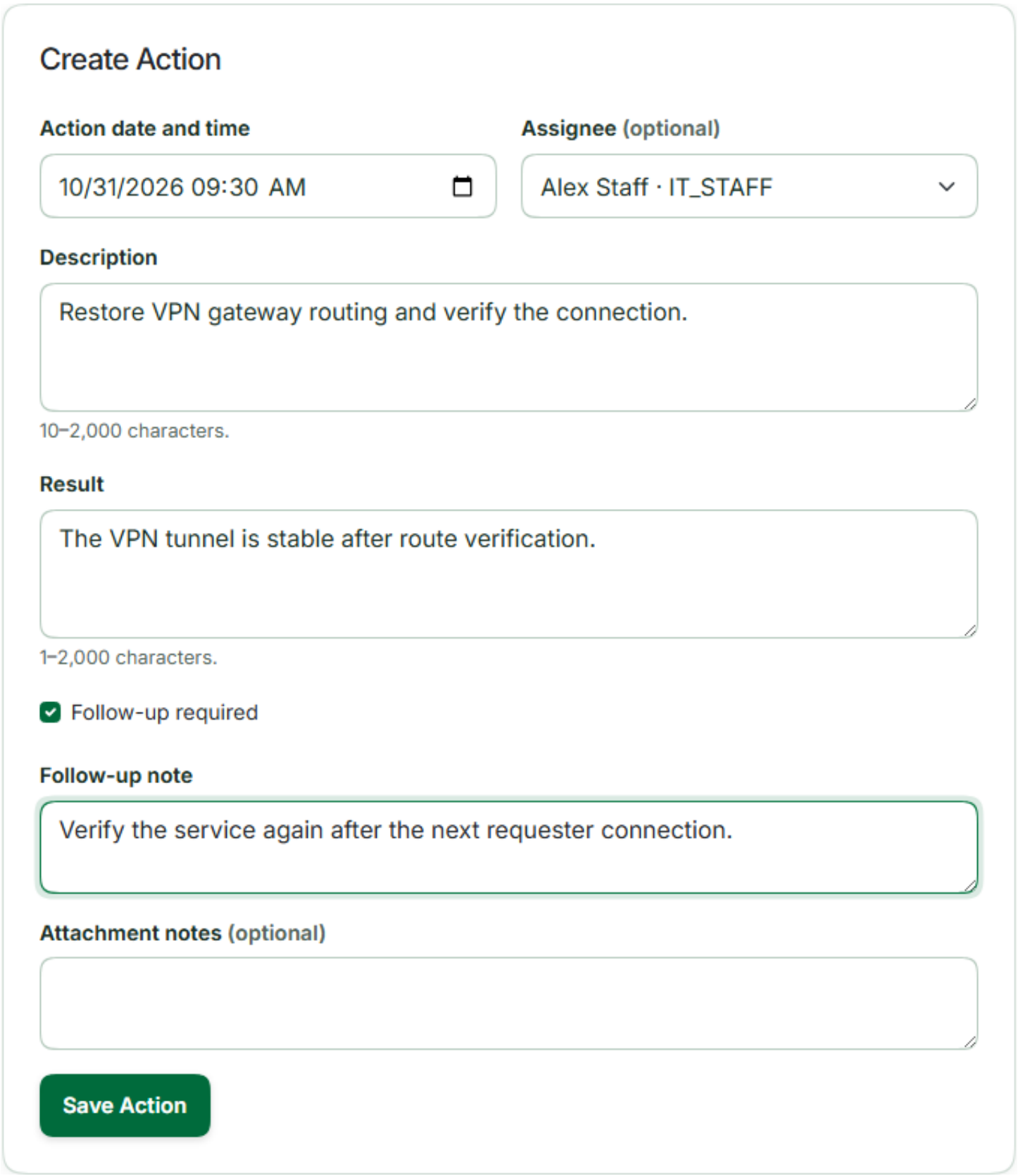
A separate authenticated API request attempted to assign an Action to an inactive IT Staff account. The server returned HTTP 400 with field error: “Assignee must be an active IT Staff member or Administrator”. The Action count stayed at 1, confirming no partial mutation.

Required-field validation



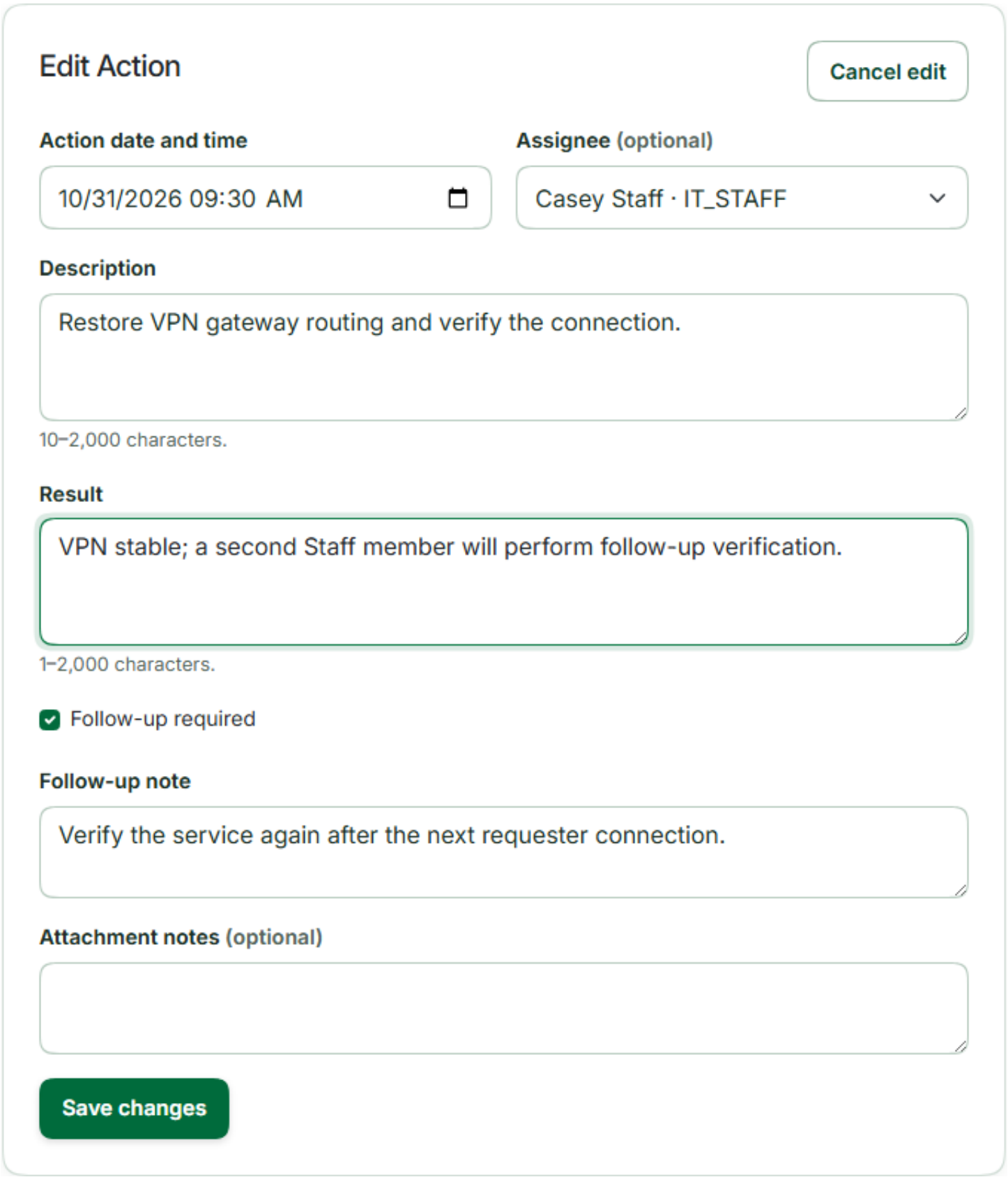
Empty save shows field errors and does not create an Action.

Action with assignee and follow-up



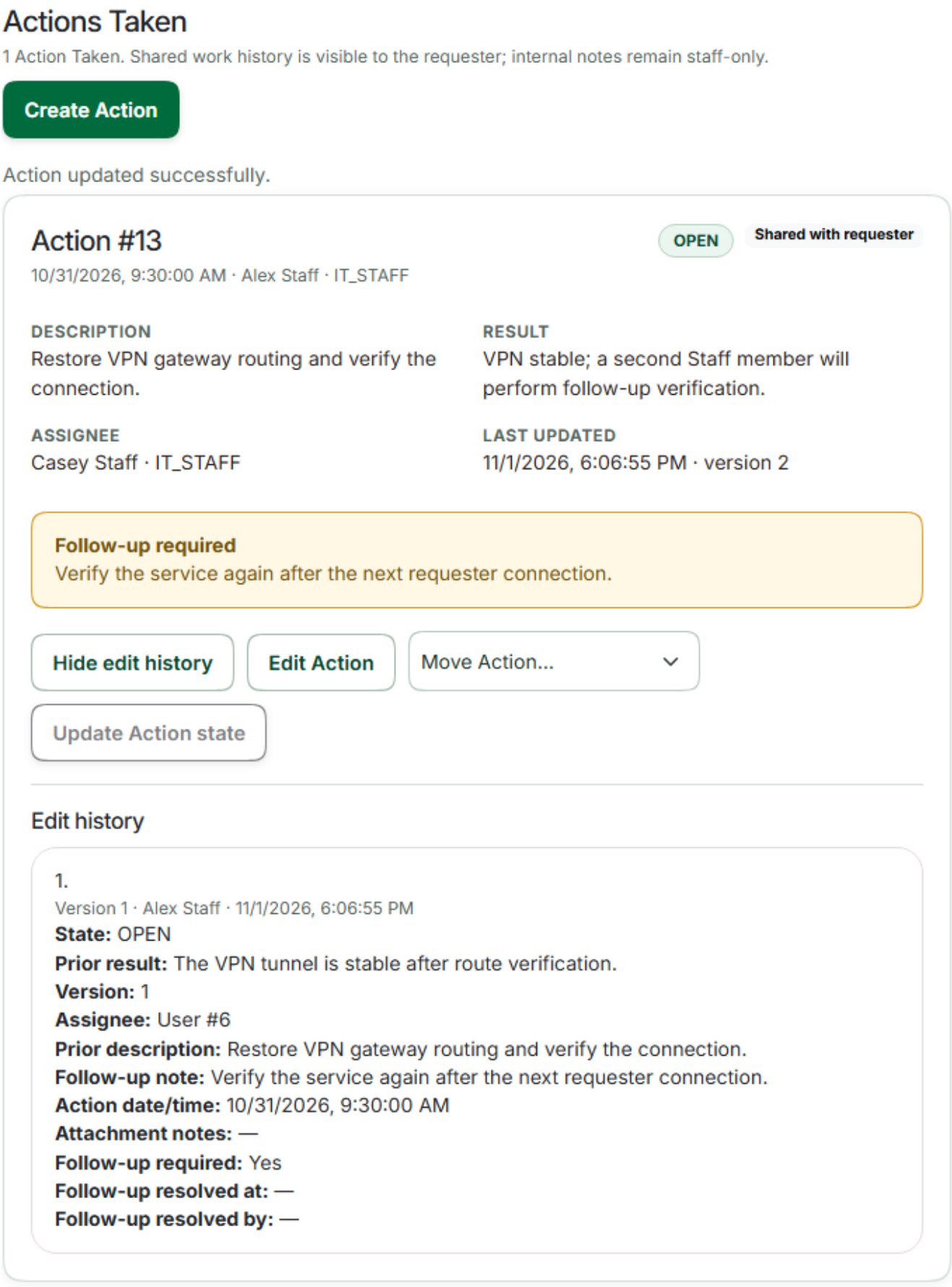
Real filled form; active Staff assignee and required follow-up note.

Edit and reassign



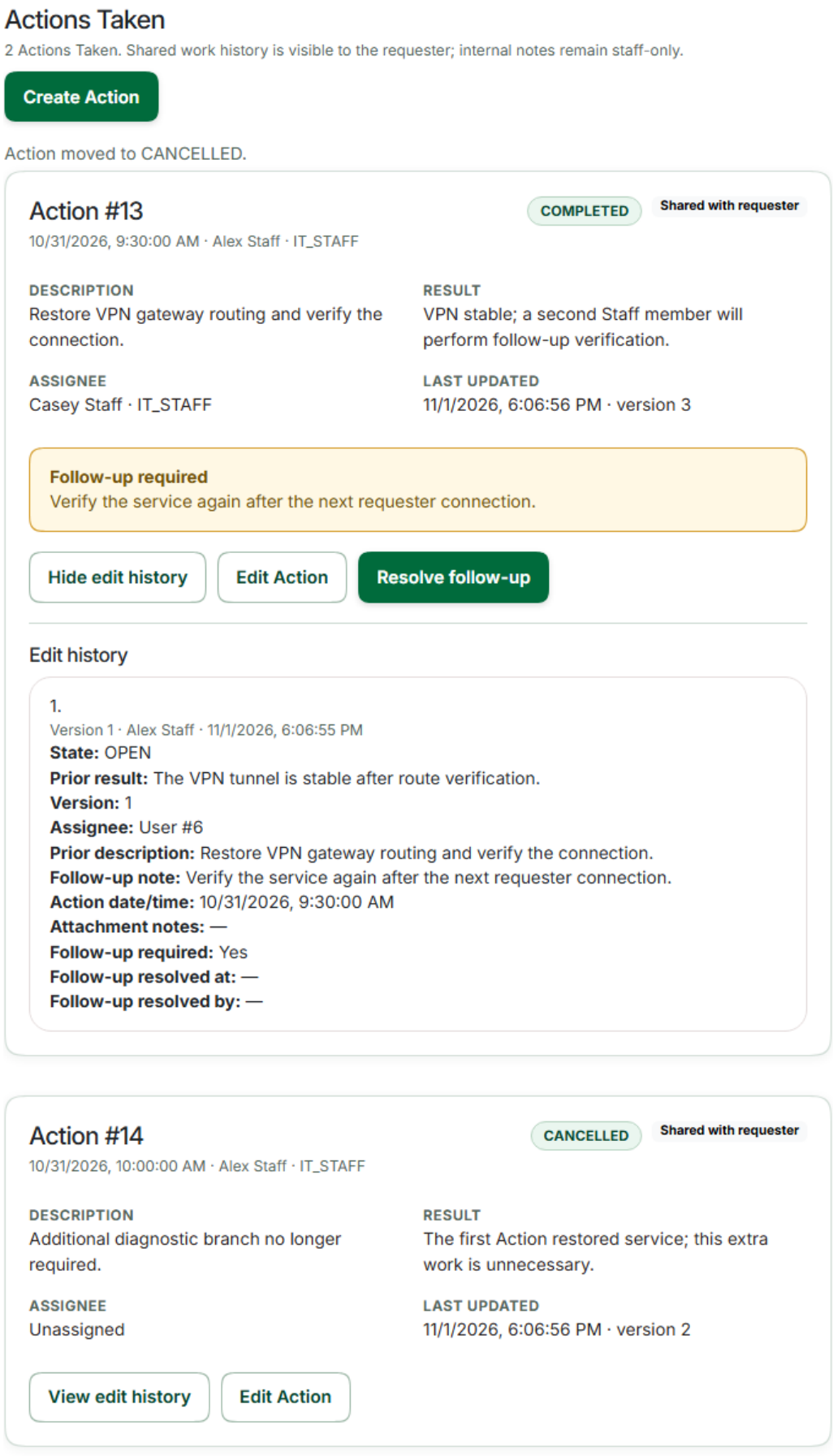
Current Staff editor changes assignee and result before save.

Immutable revision timeline



Saved prior result, assignee and version remain visible in the history.

Multiple Action states



The Ticket contains one COMPLETED Action and a second CANCELLED Action.

Requester read-only history

Actions Taken

2 Actions Taken. Shared work history is visible to the requester; internal notes remain staff-only.

Action #13

COMPLETED · Shared with requester

10/31/2026, 9:30:00 AM · Alex Staff · IT_STAFF

DESCRIPTION
Restore VPN gateway routing and verify the connection.

RESULT
VPN stable; a second Staff member will perform follow-up verification.

ASSIGNEE
Casey Staff · IT_STAFF

LAST UPDATED
11/1/2026, 6:06:57 PM · version 4

Follow-up resolved
Verify the service again after the next requester connection.
Resolved 11/1/2026, 6:06:57 PM by Alex Staff.

View edit history

Action #14

CANCELLED · Shared with requester

10/31/2026, 10:00:00 AM · Alex Staff · IT_STAFF

DESCRIPTION
Additional diagnostic branch no longer required.

RESULT
The first Action restored service; this extra work is unnecessary.

ASSIGNEE
Unassigned

LAST UPDATED
11/1/2026, 6:06:56 PM · version 2

View edit history

Requester sees both Actions; create/edit and internal-note controls are absent.

6.1 Field-level and lifecycle evidence

Case	Observed result
Required-field validation	Empty description/date/result is rejected inline; no Action row is created.
Active assignment + follow-up	Active Staff assignee is allowed; a note is required when follow-up is selected.
Inactive assignment	HTTP 400 field error; Action count remained 1.
Edit / audit	Accepted edit creates a prior-value revision; creation and performer audit remain visible.
Action states	A COMPLETED Action and a CANCELLED Action are both shown; terminal controls are unavailable.
Requester view	Both Actions are visible, with no create/edit controls and no internal-note controls.

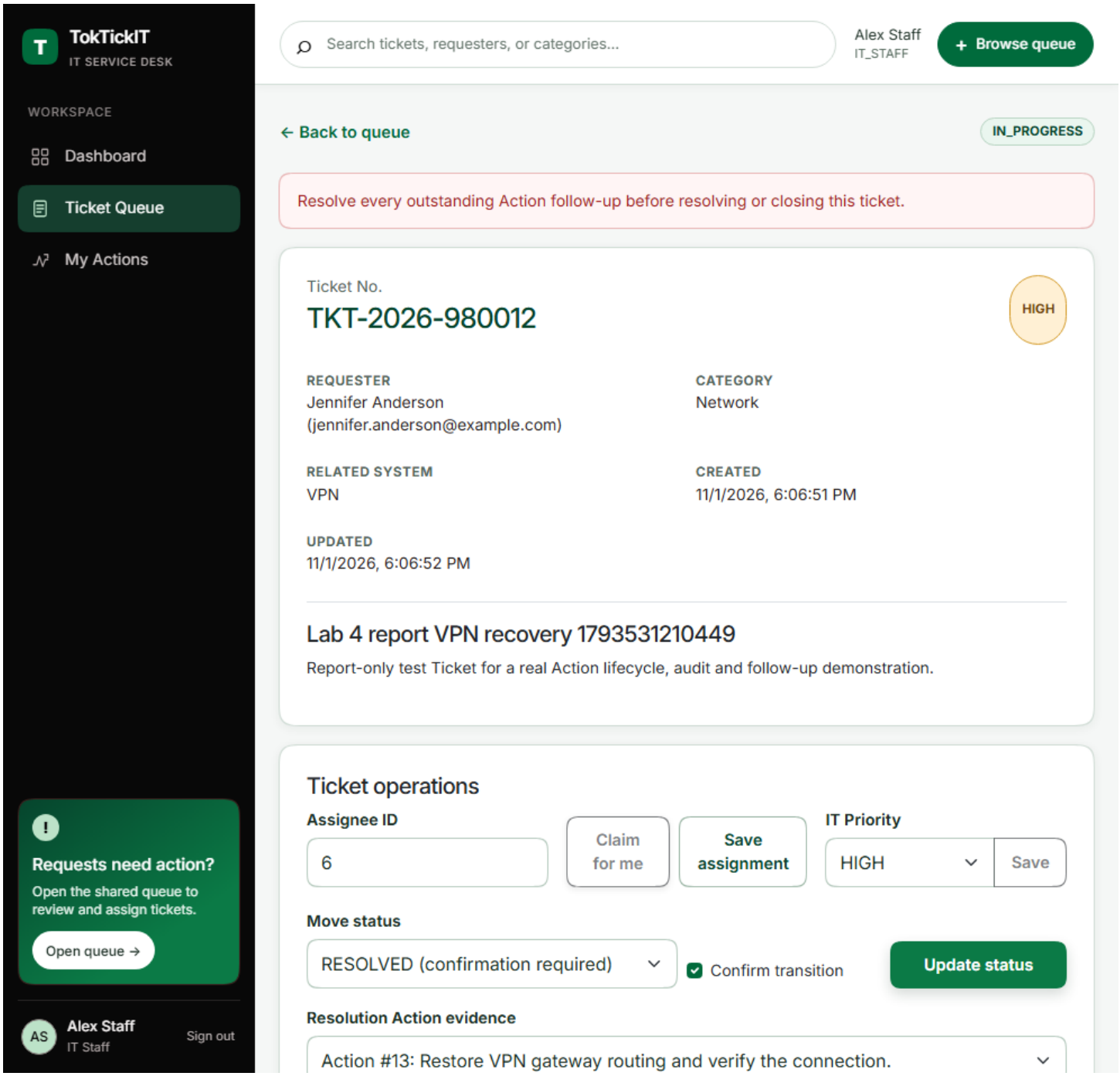
Actions API and field contracts

Answer Part 7: Ticket Workflow and Resolution Gate (5 Points)

The server requires a completed qualifying Action and no unresolved follow-up before RESOLVED or CLOSED. An unresolved follow-up blocks resolution even when the Action itself is COMPLETED. Authorized Staff/Admin can resolve the follow-up using its current version; that appends an audit revision while preserving the Action completion state. Reopen follows the defined confirmation rule.

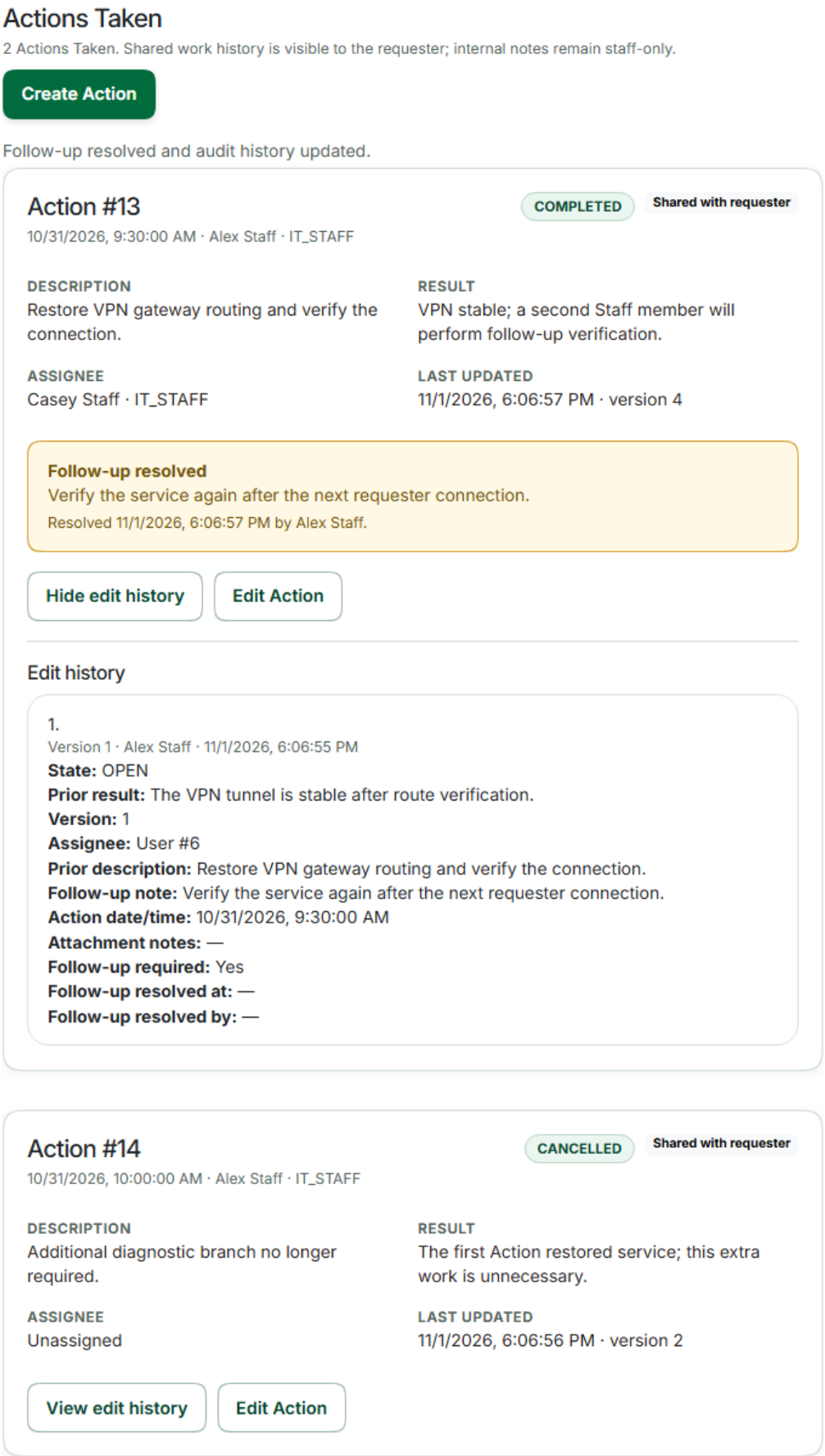
The browser flow recorded Ticket 12 on the local test database. A resolution attempt was blocked with the actual server message: “Resolve every outstanding Action follow-up before resolving or closing this ticket.”. After follow-up resolution, the Ticket transitioned to RESOLVED and then REOPENED. These are the actual tested transitions; they do not indicate a change to the production database.

Resolution blocked by outstanding follow-up



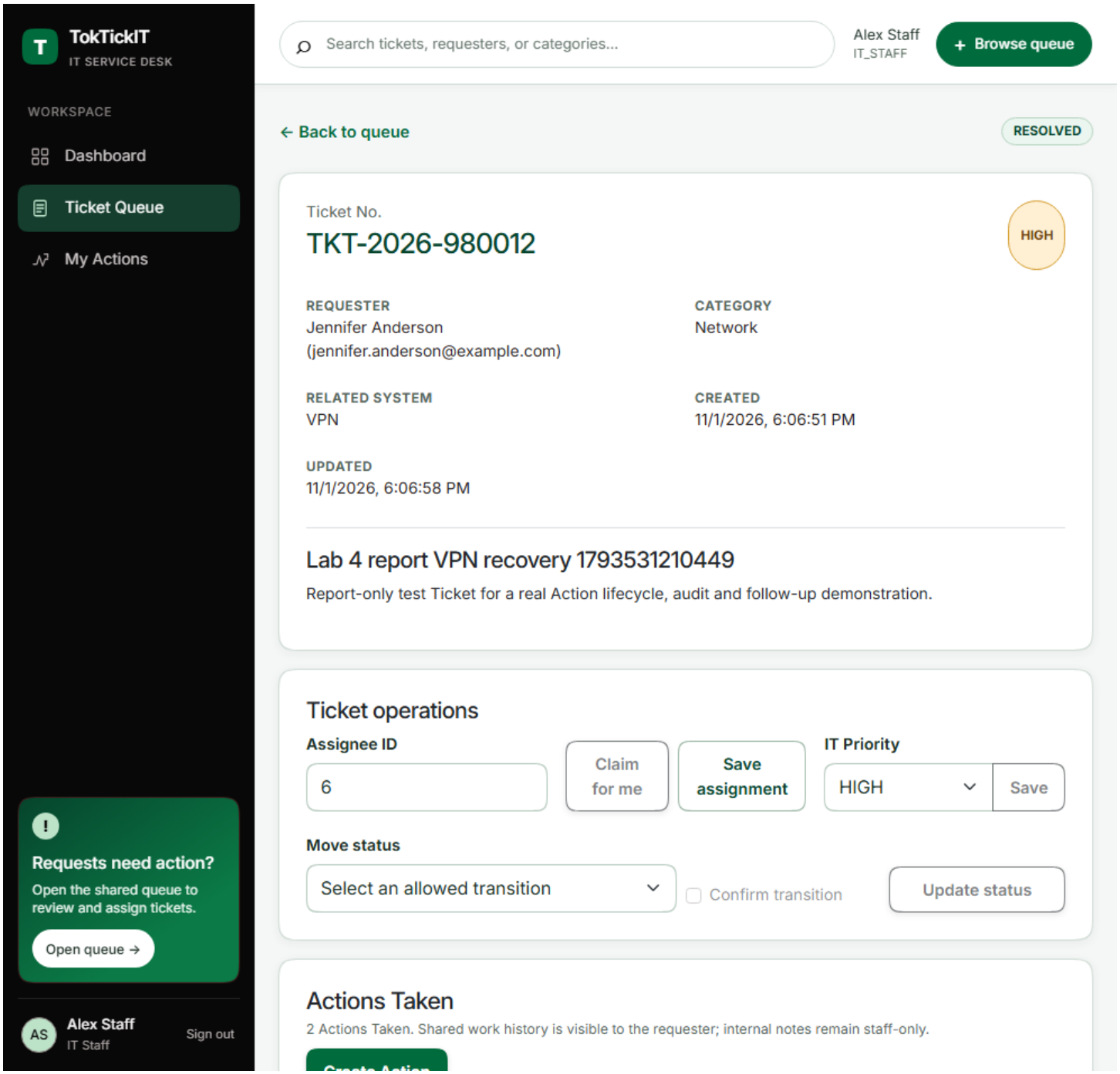
Ticket remains IN_PROGRESS; exact server gate message is stated above.

Follow-up resolved



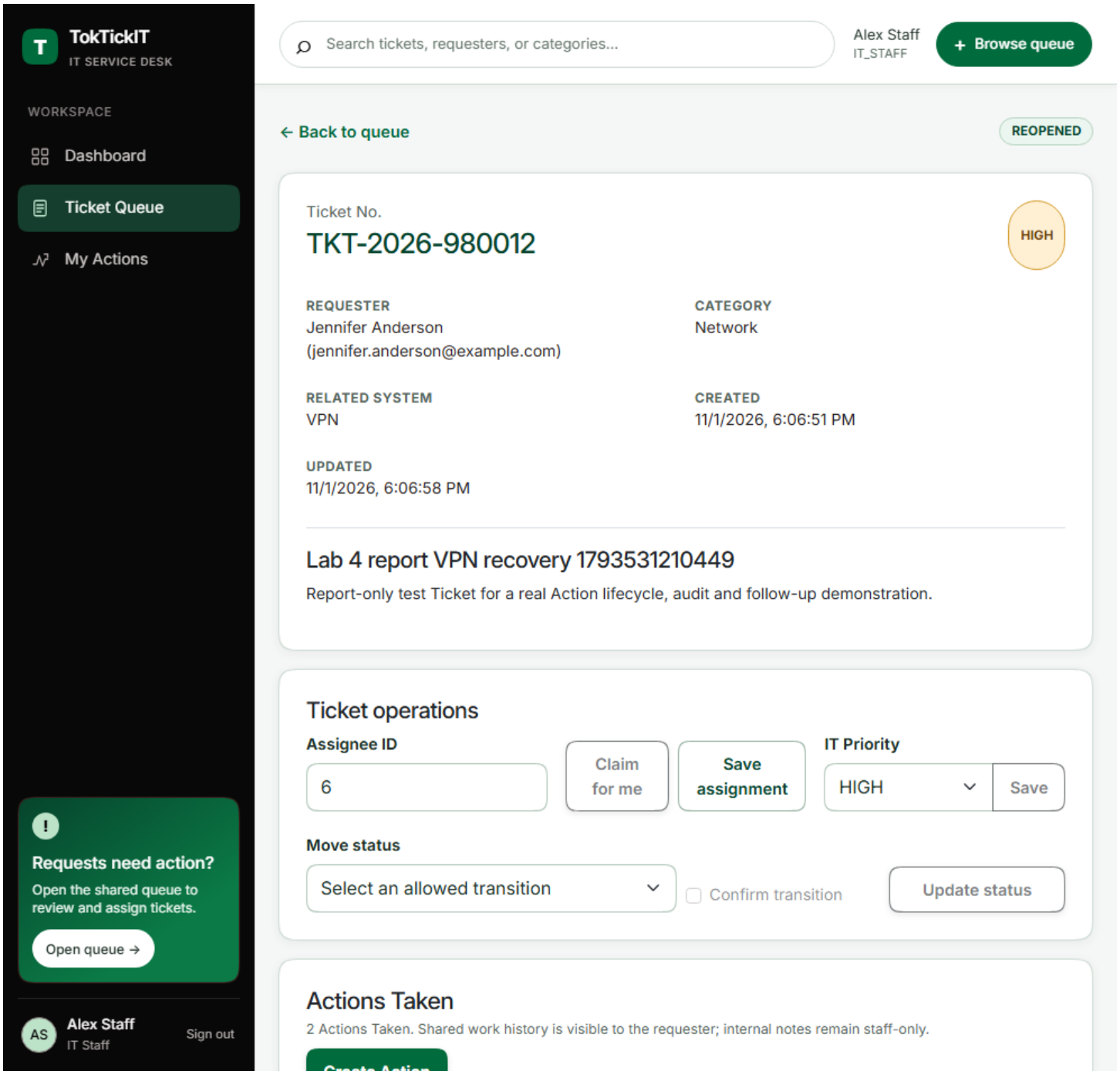
Authorized Staff resolves the follow-up; Action remains COMPLETED.

Ticket formally resolved



After clearing the follow-up, Staff resolves the Ticket.

Reopen from RESOLVED

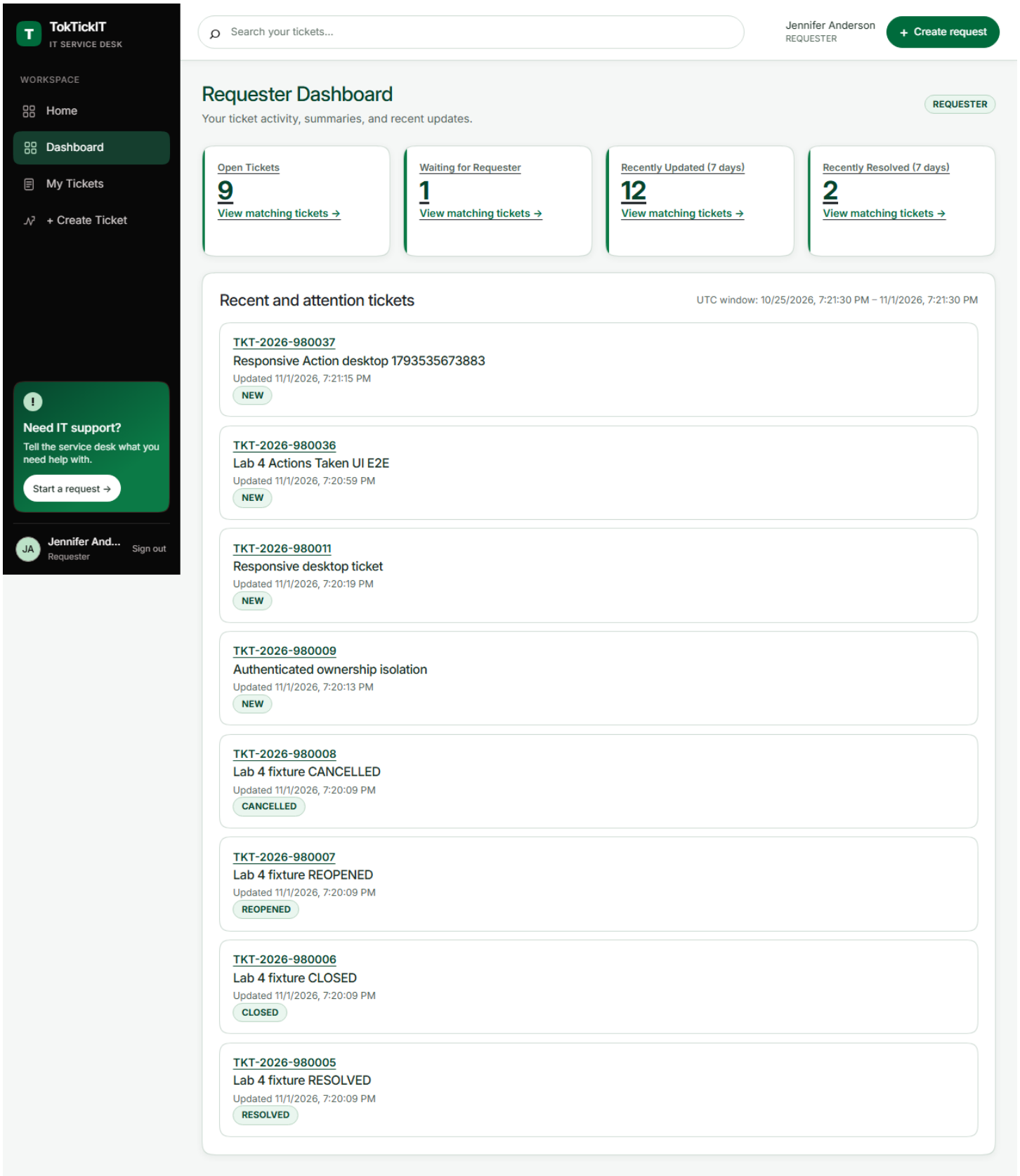


The reopened Ticket and permitted next-step controls are visible.

Answer Part 8: Requester Dashboard and Lab 1–3 Regression (5 Points)

The full regression suites retain the approved Lab 1–3 contracts: ticket creation and ownership, attachment rules, requester isolation, authentication/password gate/logout, role authorization, comments/notes, Staff queue/detail operations and Administrator user management. The refreshed responsive release views below were captured in the post-merge E2E run; Lab 4 dashboard, Actions and workflow examples appear in Parts 5–7.

Requester Dashboard · desktop



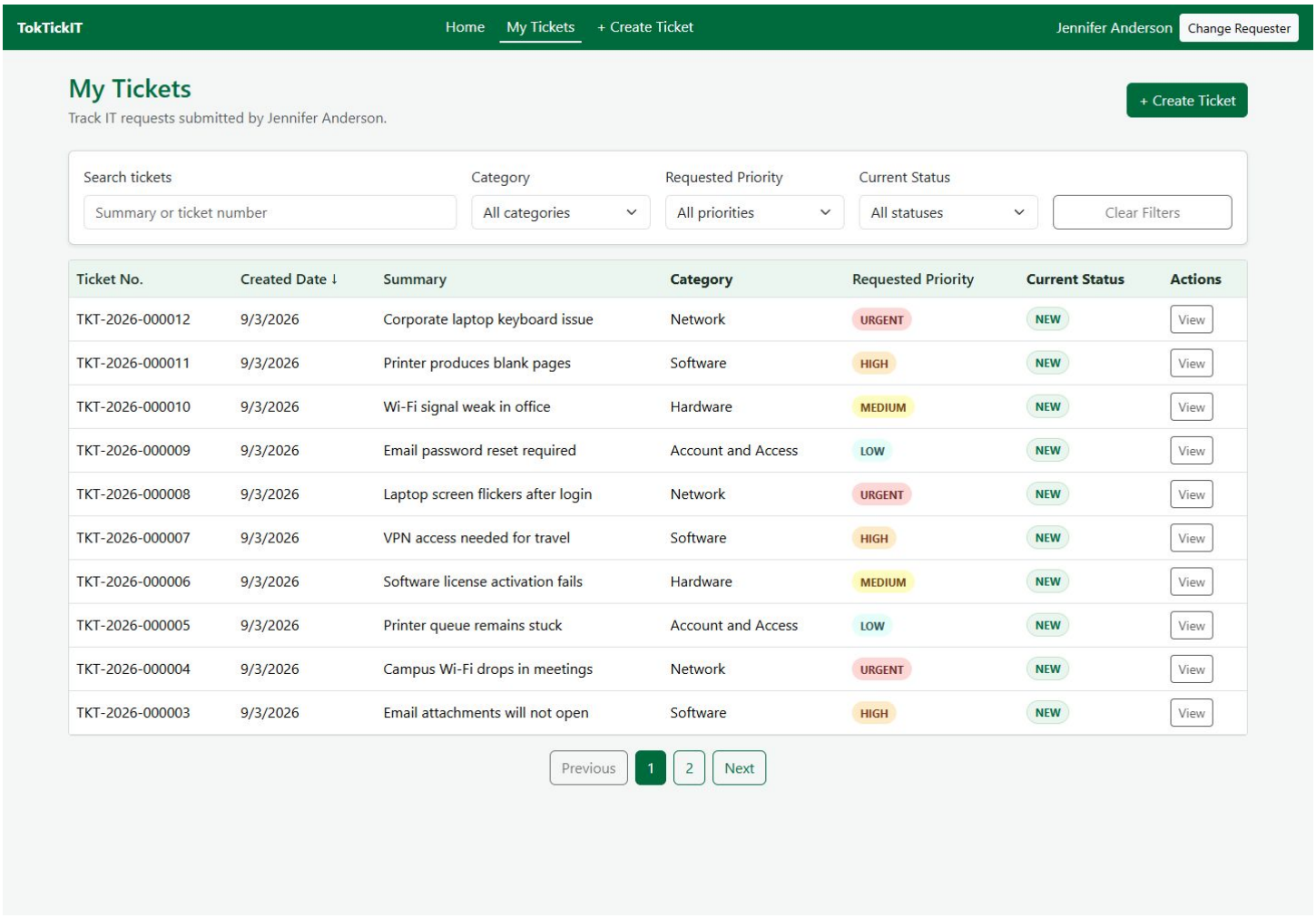
Fresh authenticated requester view after PR #64 merged; dashboard counts and ticket links are scoped to the signed-in requester.

8.1 Regression and state coverage

Area	Evidence preserved
Lab 1–2 ticket journeys	Authenticated creation, My Tickets, detail, attachment allow/deny, safe ownership and requester isolation.
Lab 3 authentication	Valid/invalid login, inactive account, first-login password change, role-aware navigation, logout and direct-access protection.
Lab 3 staff operations	Queue sorting/filtering/pagination, claim/reassign, status confirmation, public comment/internal note and forbidden access.
Lab 3 Administrator	User list/search, create/edit, role/status changes, password reset and non-admin/last-admin protections.
Lab 4 regression	Actions, dashboard metrics/drill-downs, status transitions, stale updates, resolution gate, retry and responsive states.

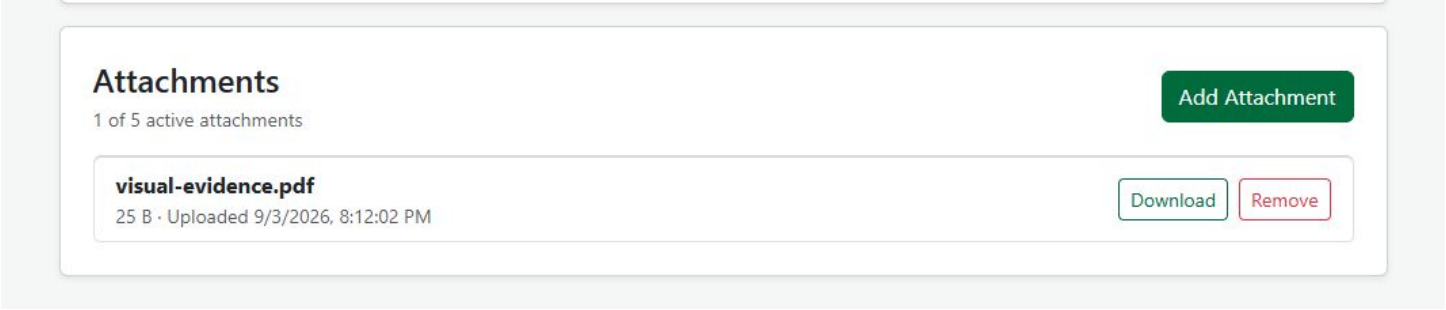
8.2 Inherited requester, communication and Administrator screens

Lab 2 · authenticated My Tickets



Real browser evidence of the signed-in requester's owned ticket list; maintained as Lab 4 regression coverage.

Lab 2 · Ticket Detail attachments



Real browser evidence of the attachment list and permitted download/removal controls.

Lab 3 · public communication



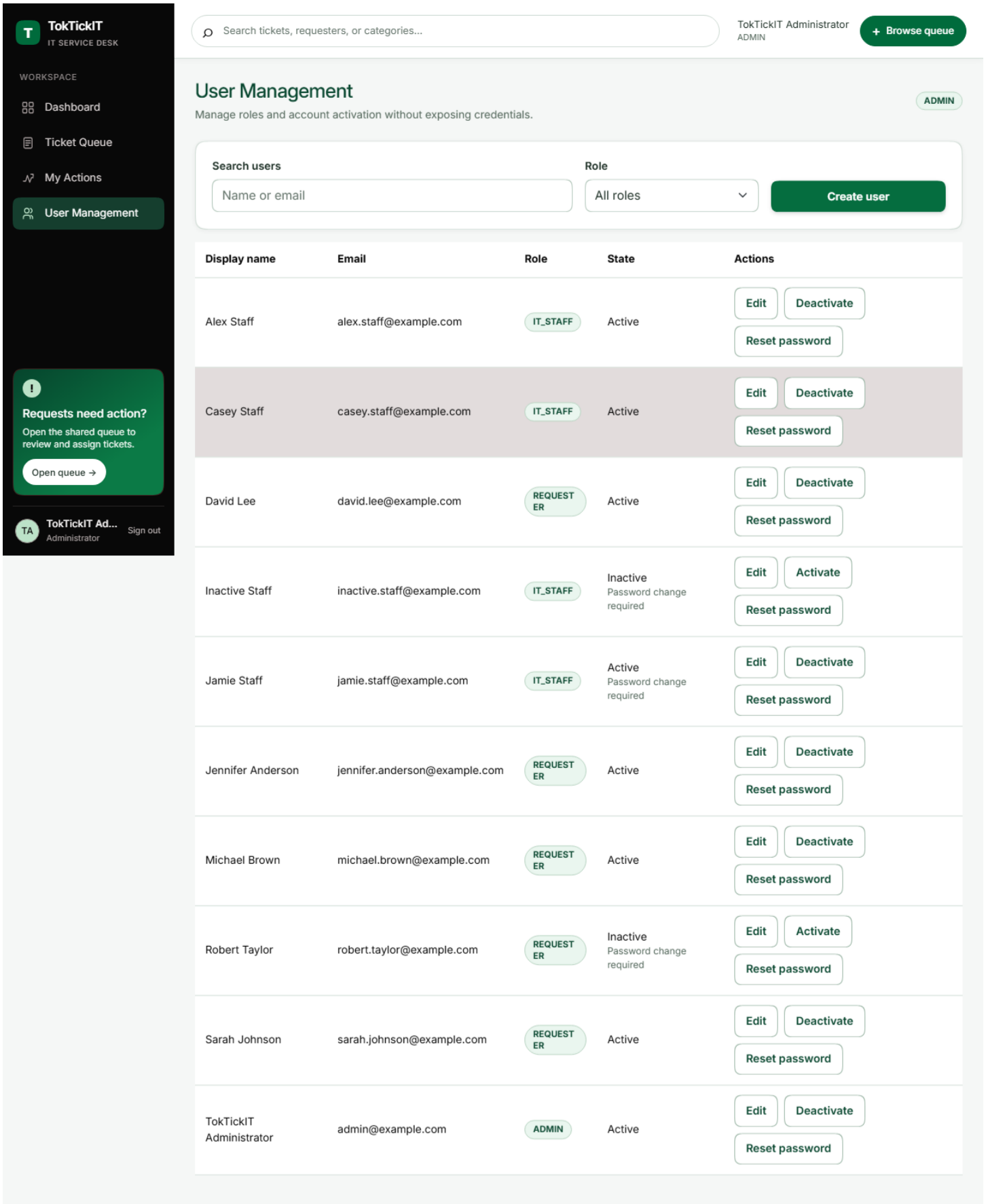
Staff posts a public response visible in the requester conversation.

Lab 3 · staff-only internal note



The internal note is labeled staff-only and remains outside the requester conversation.

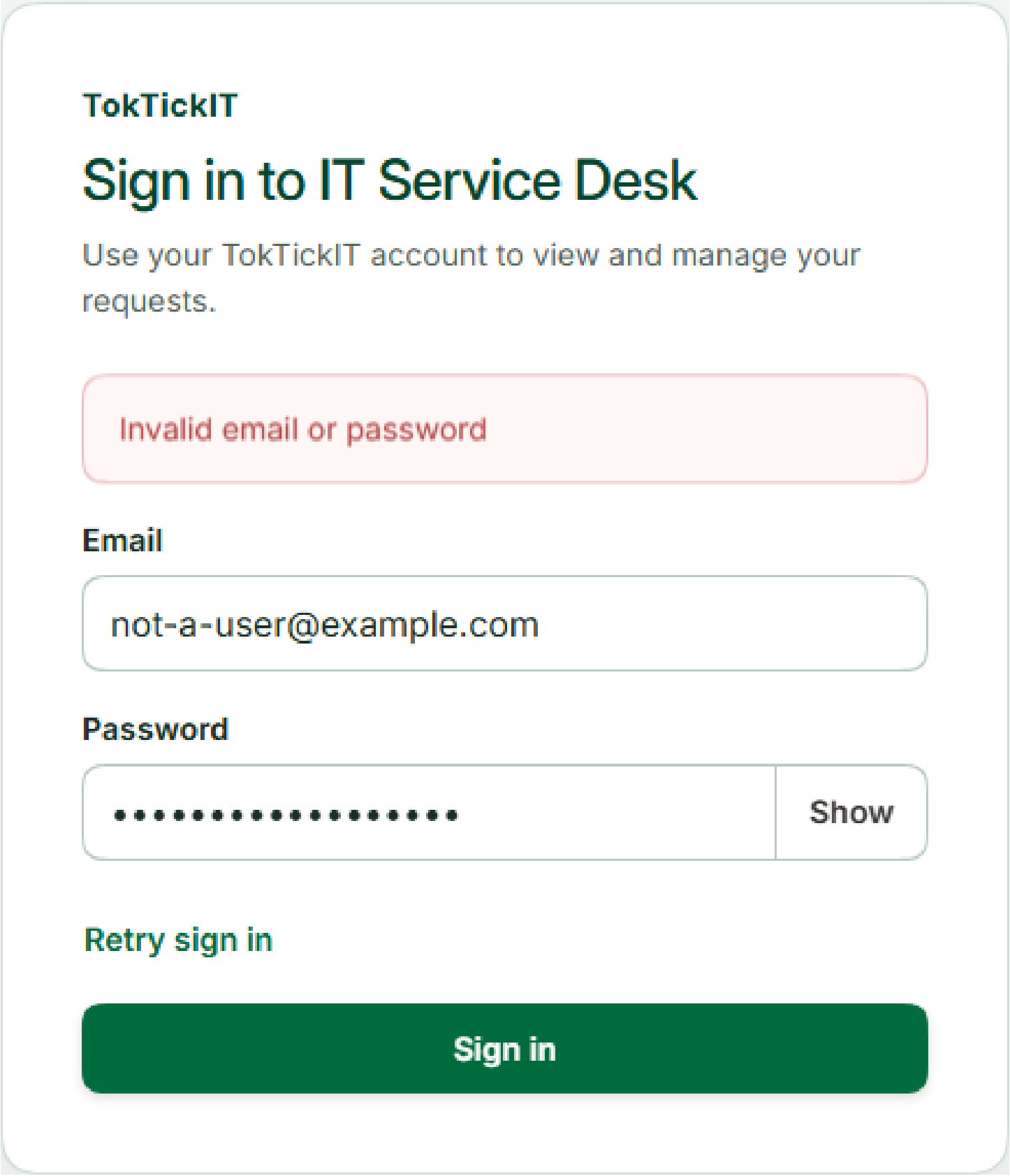
Lab 3 · Administrator user management



Authenticated Administrator view of searchable/filterable user accounts.

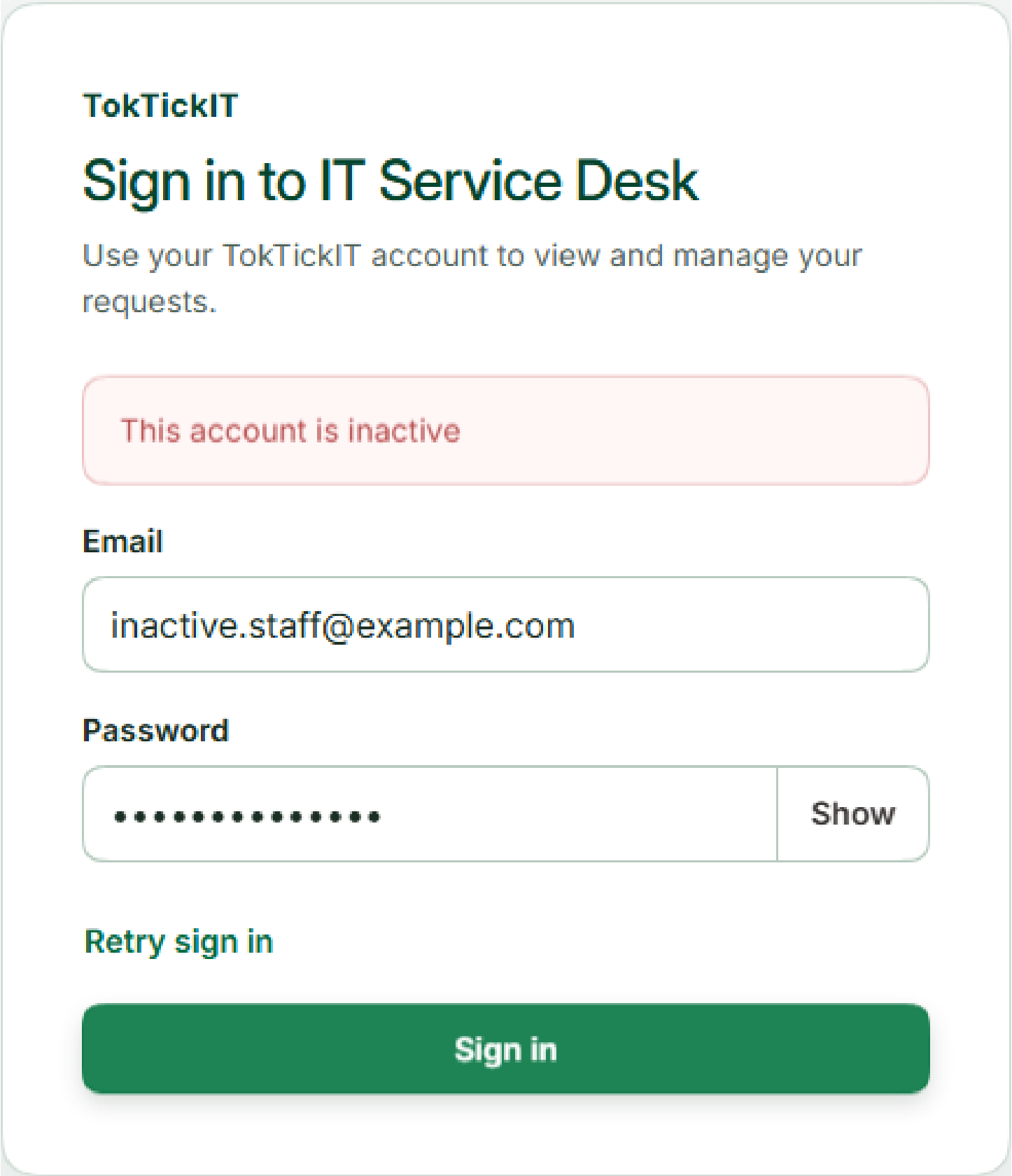
8.3 Existing authentication state evidence

Invalid password; generic sign-in error.



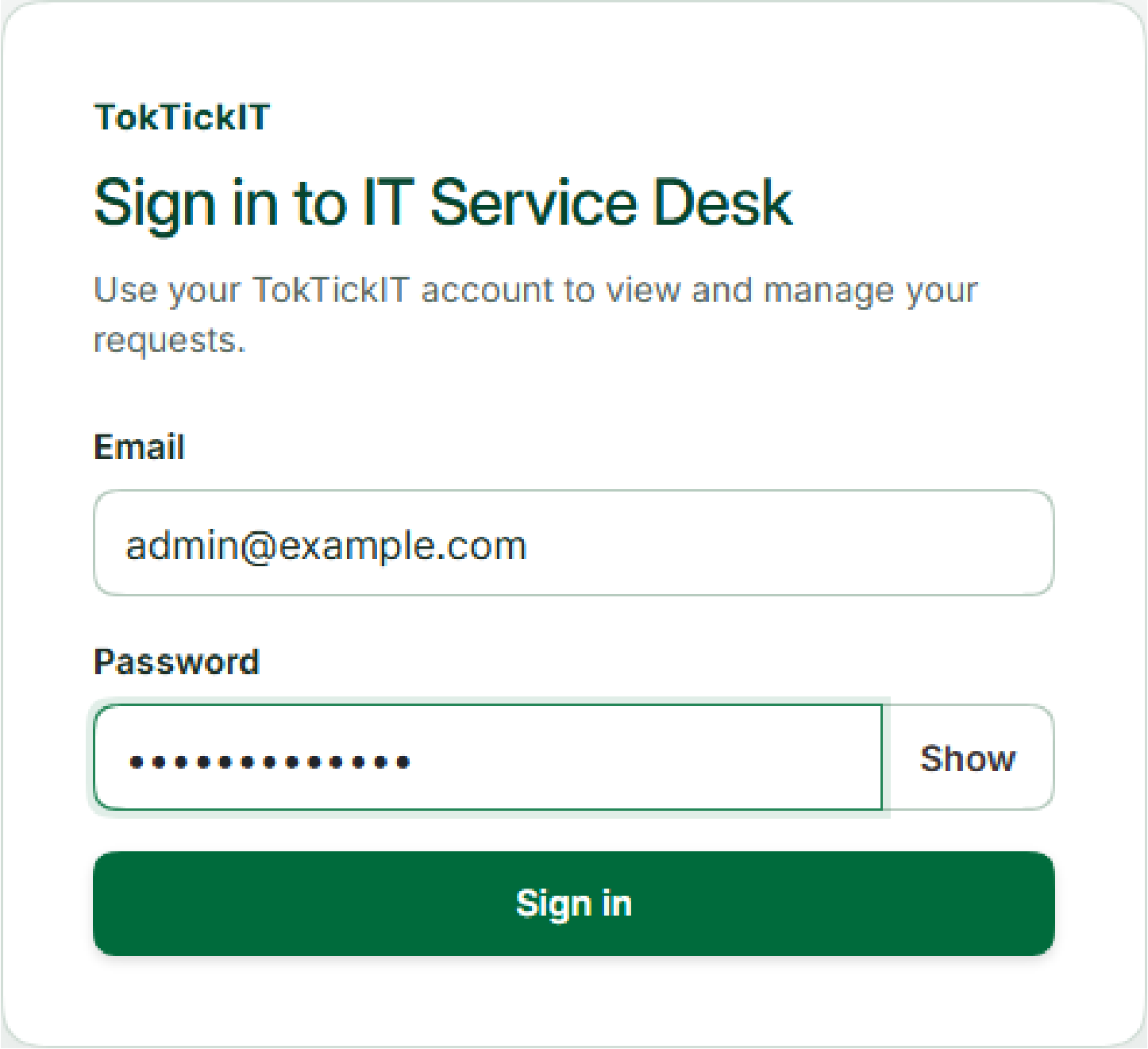
Real browser evidence for the inherited Lab 3 authentication flow.

Inactive seeded account response.



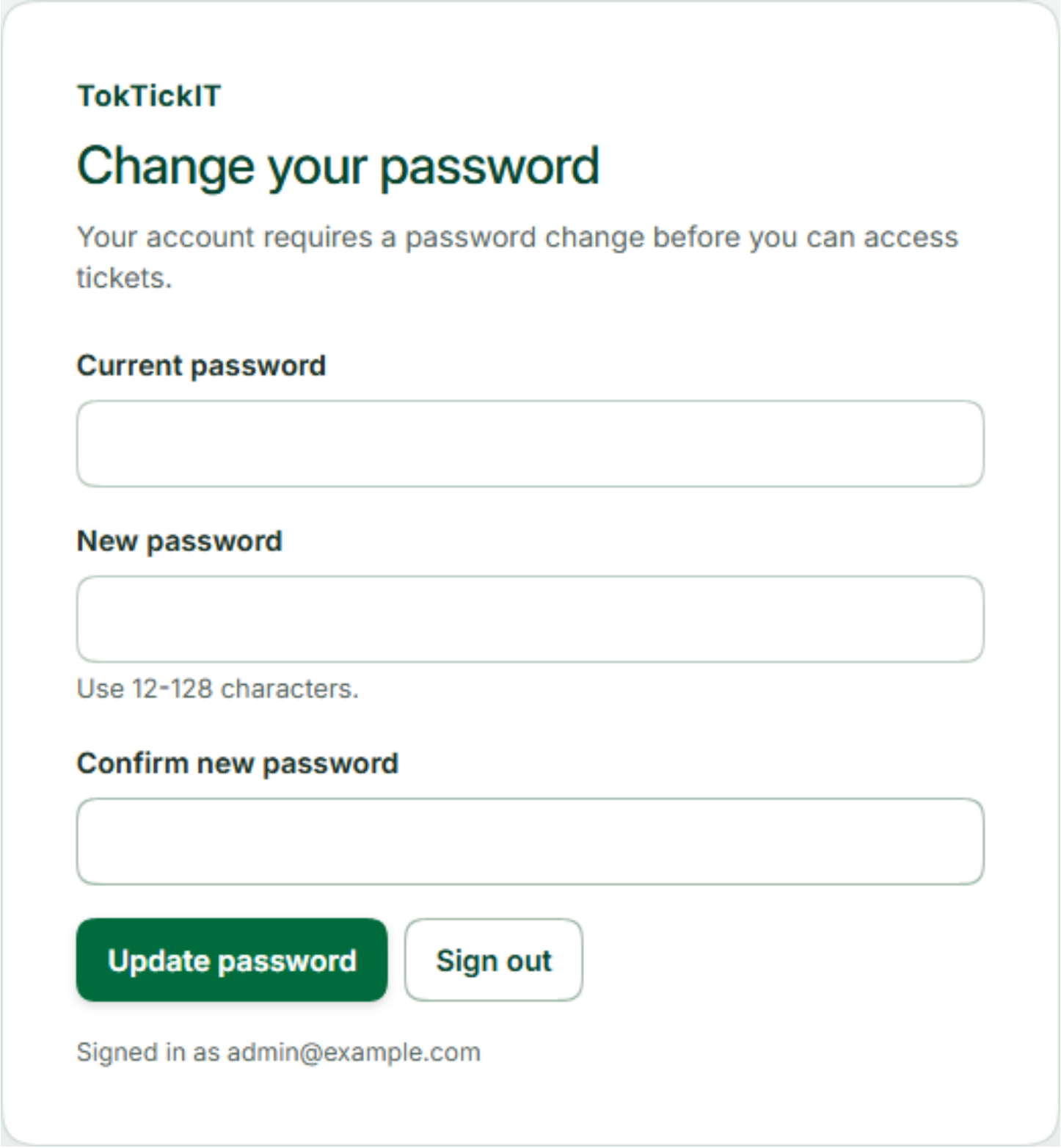
Real browser evidence for the inherited Lab 3 authentication flow.

Valid credentials submitted through the UI.



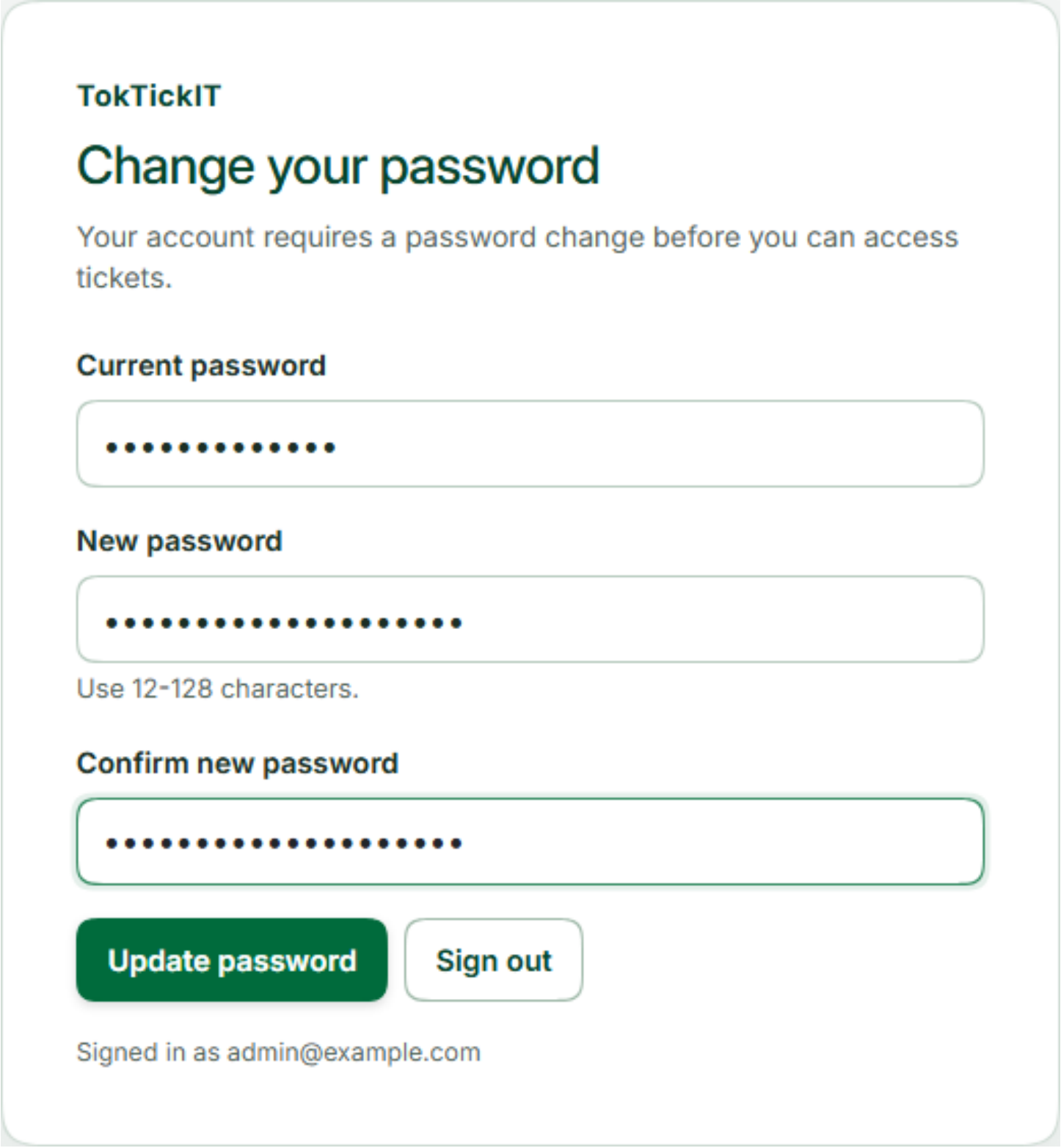
Real browser evidence for the inherited Lab 3 authentication flow.

First-login password gate takes precedence.



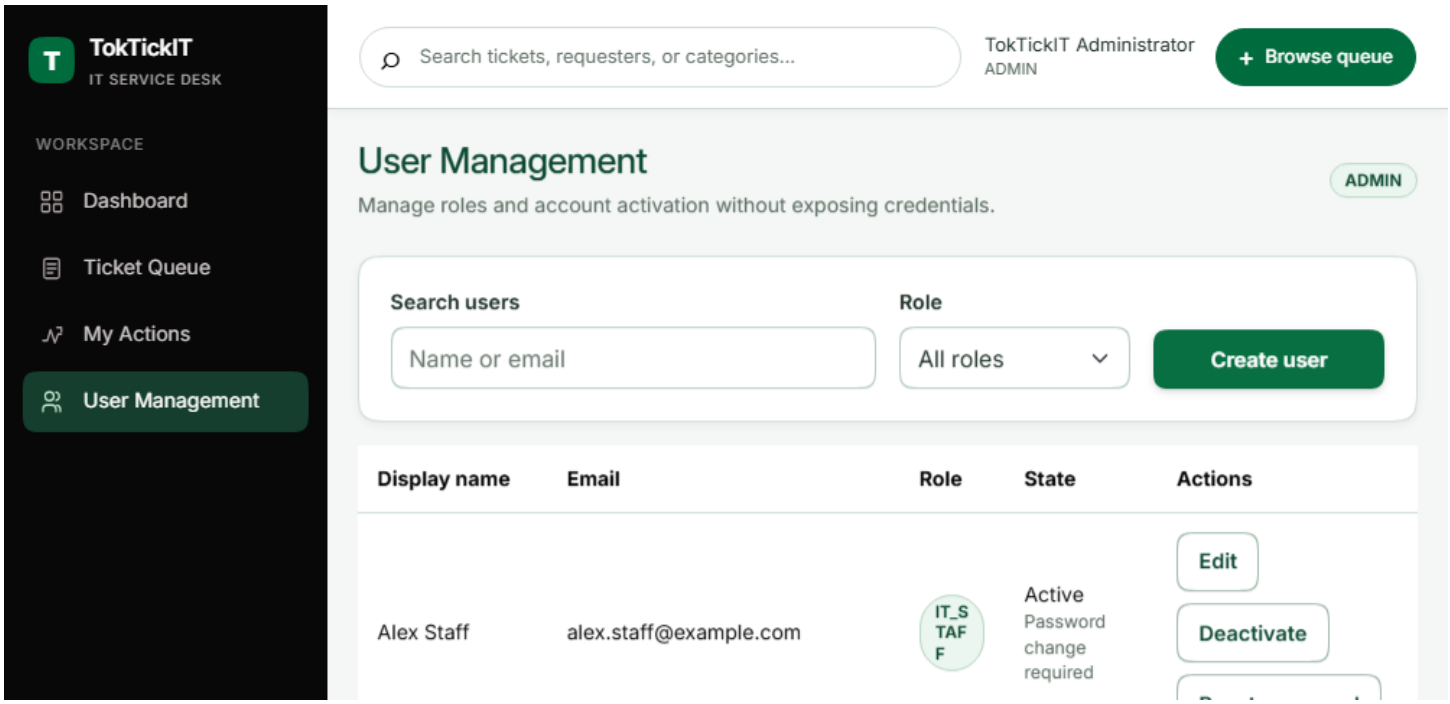
Real browser evidence for the inherited Lab 3 authentication flow.

First-login password change form



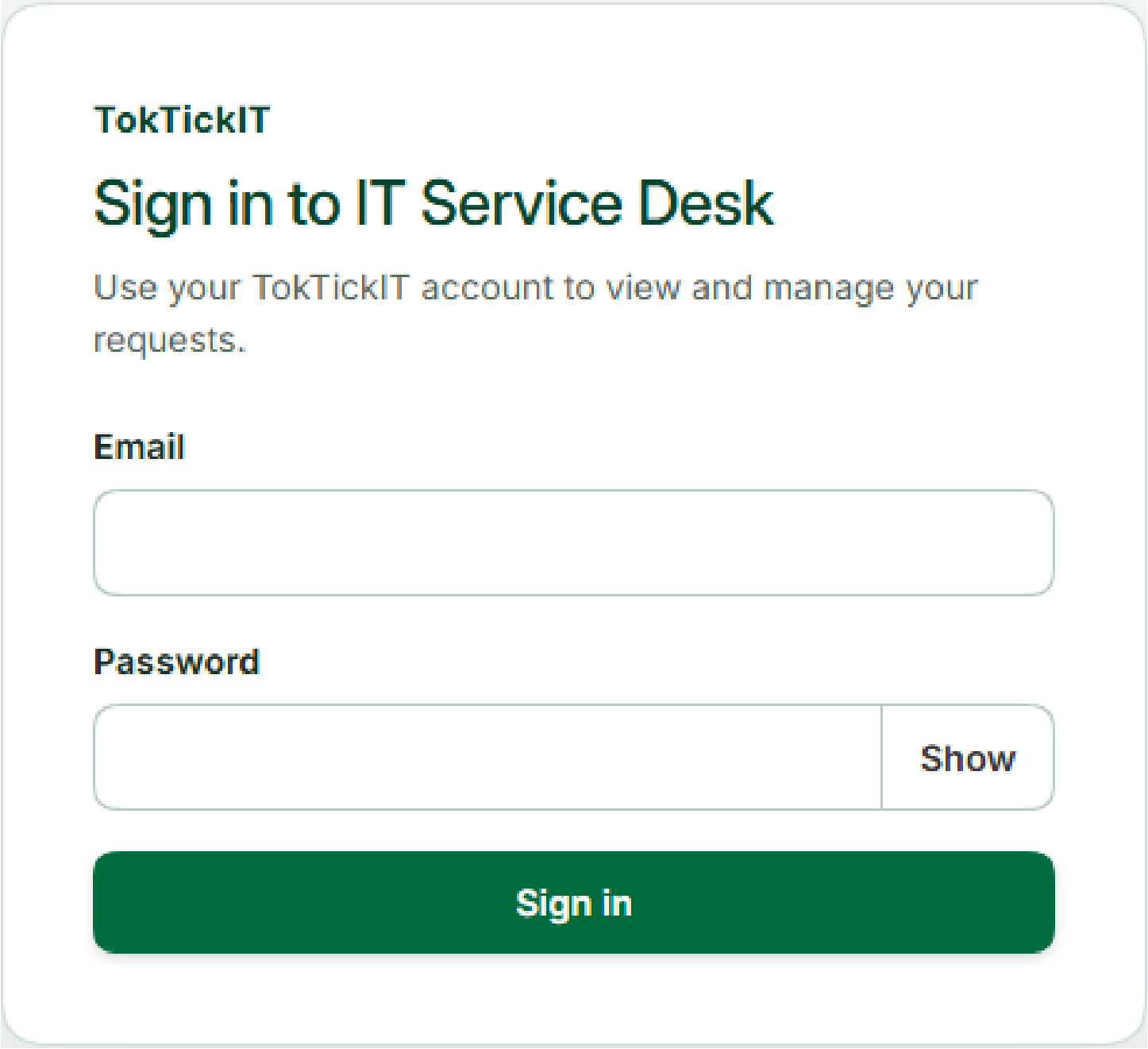
Fresh authenticated E2E capture after PR #64 merged to main.

Password change and sign-in success



Fresh authenticated E2E capture after PR #64 merged to main.

Protected route after logout

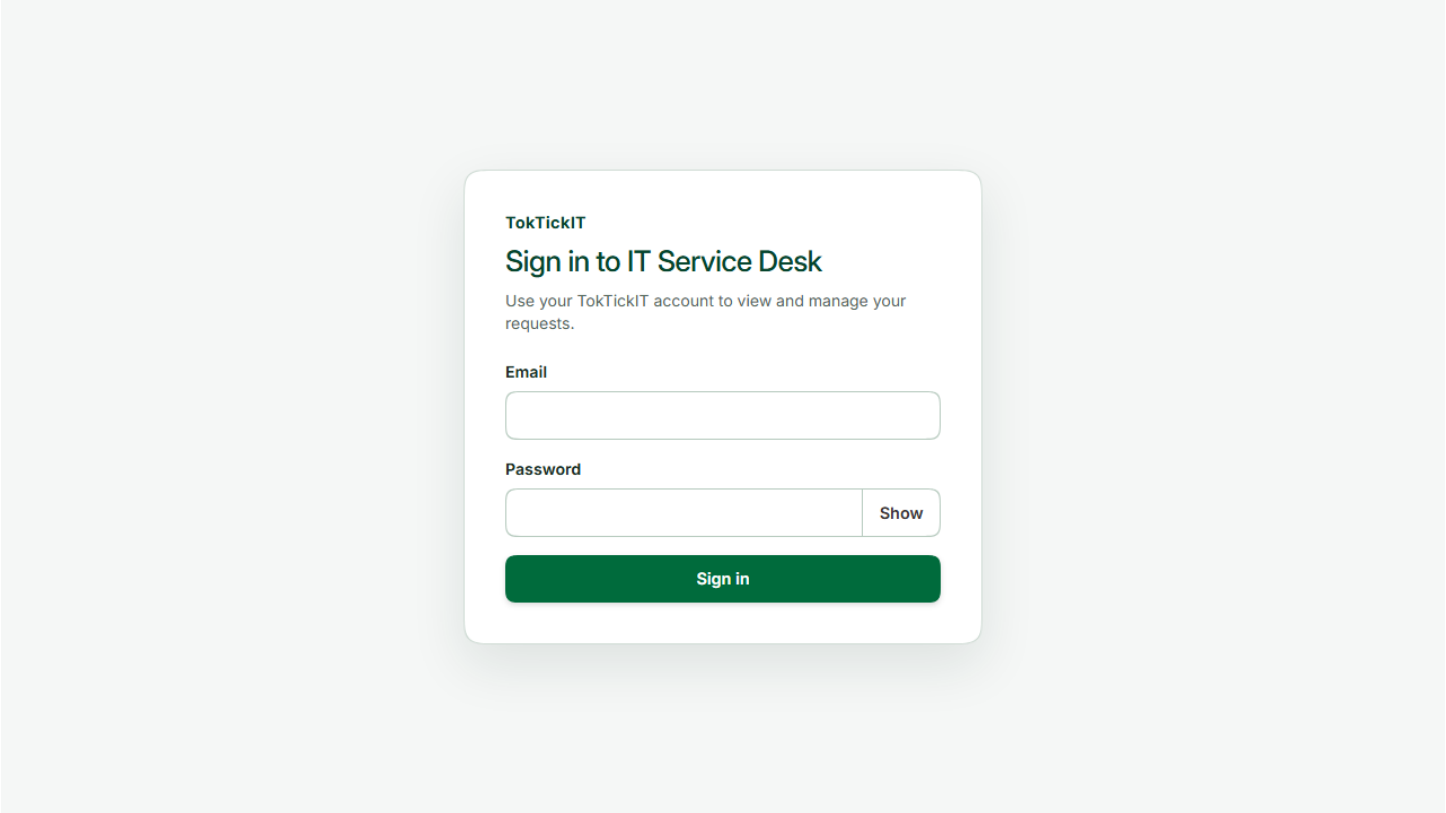


Fresh authenticated E2E capture after PR #64 merged to main.

8.4 Inherited protected screens

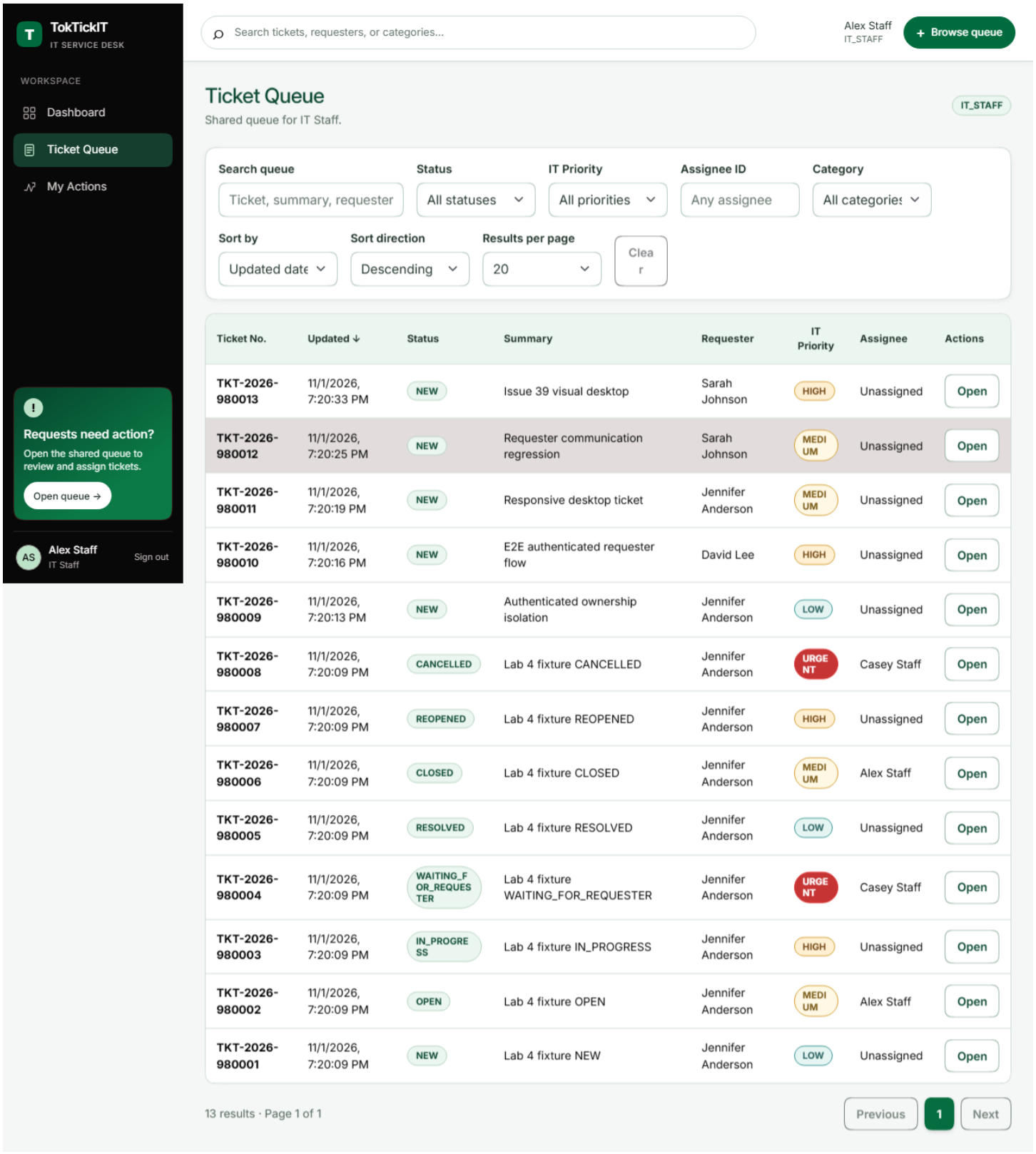
These representative Lab 3 screens are preserved as real E2E captures. The inherited responsive behavior remains covered by the named Lab 1–3 regression flows and the original screenshot set; Part 9 contains the full desktop/tablet/mobile comparison for every major Lab 4 view.

Authenticated shell · desktop



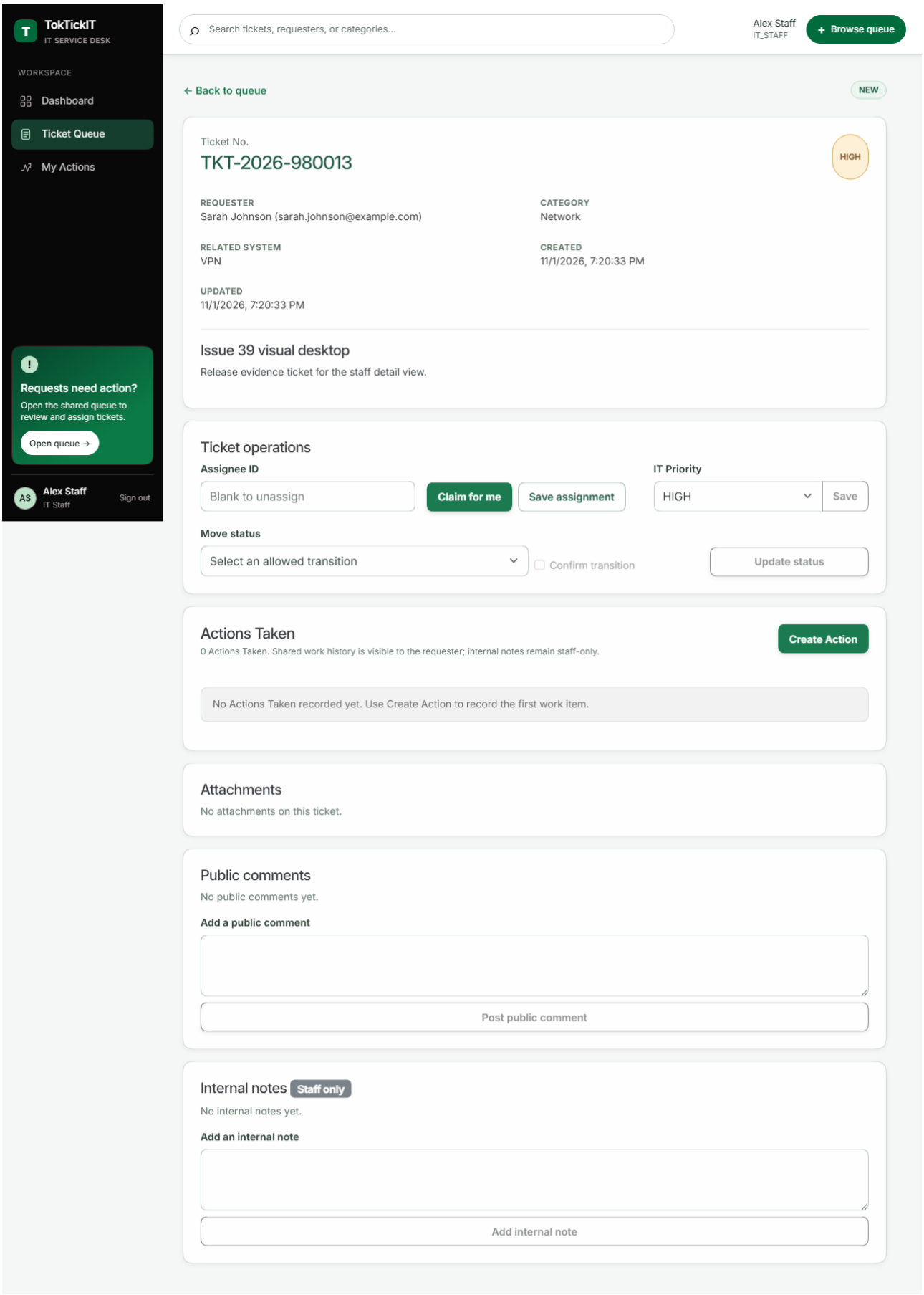
The shared session shell and role-aware navigation continue to gate protected screens.

Staff Queue · desktop



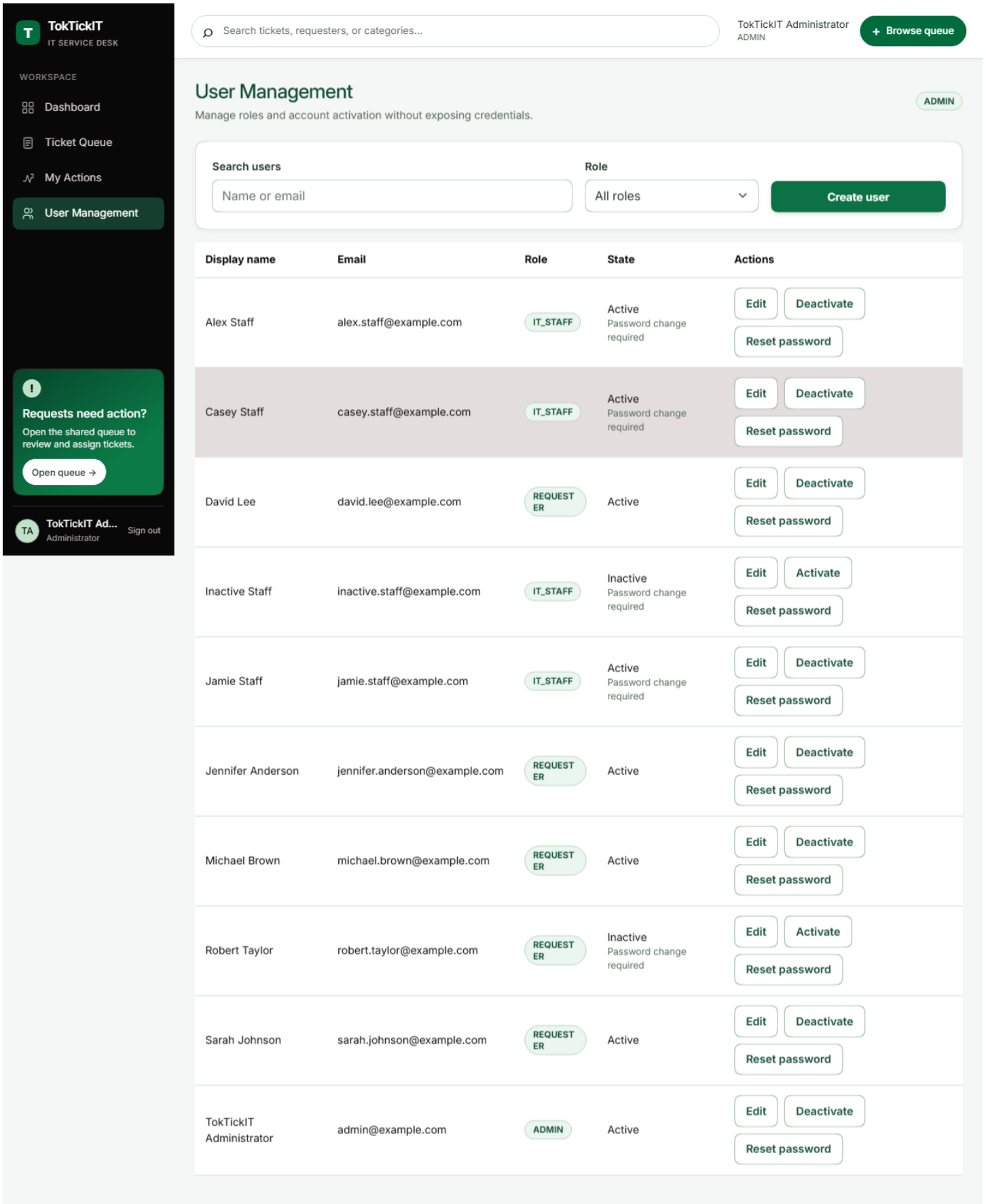
Authenticated Staff can filter and open the shared operational queue.

Staff Ticket Detail · desktop



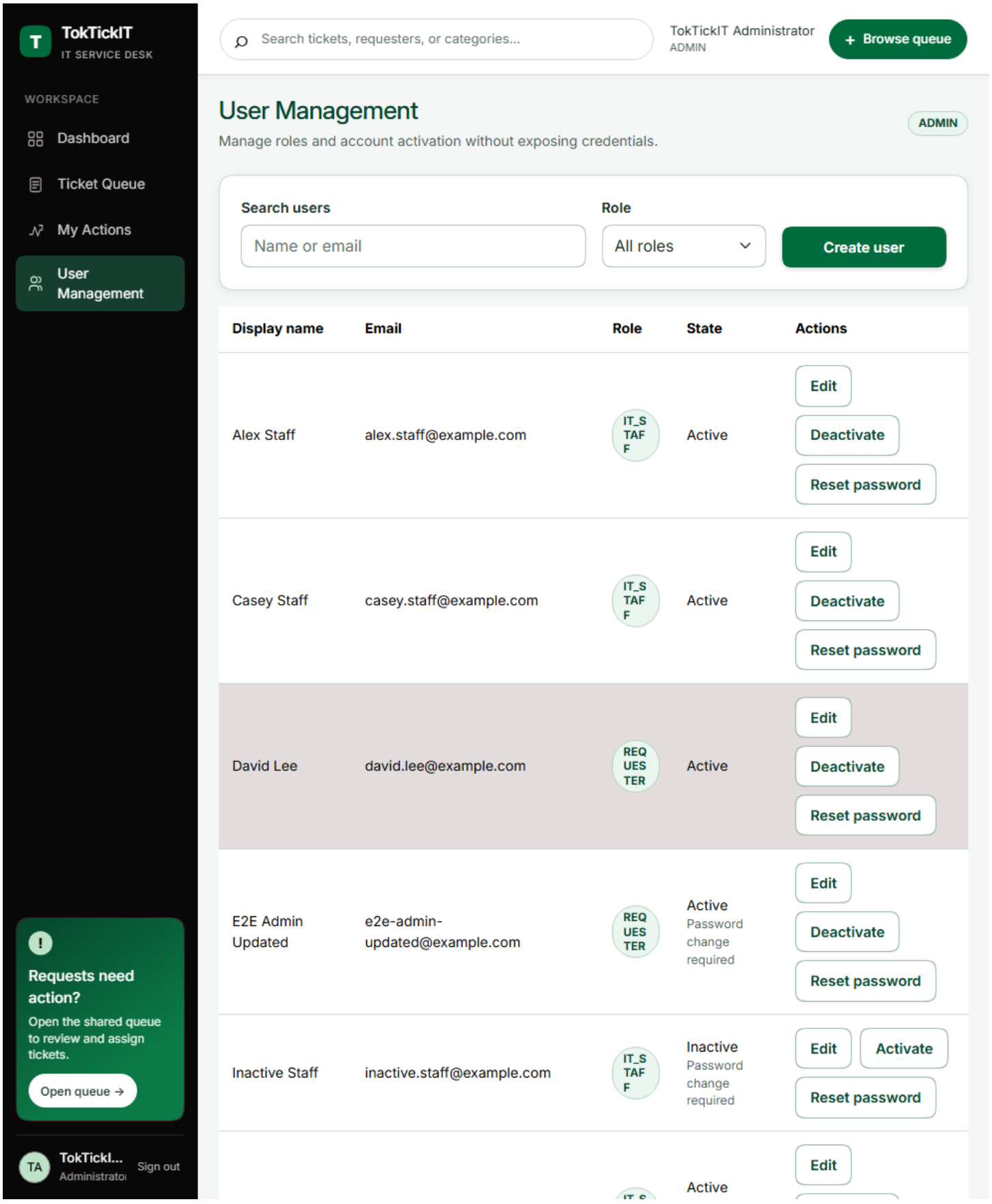
Staff sees assignment, priority, status, Actions Taken, comments and internal notes.

Administrator User Management · desktop



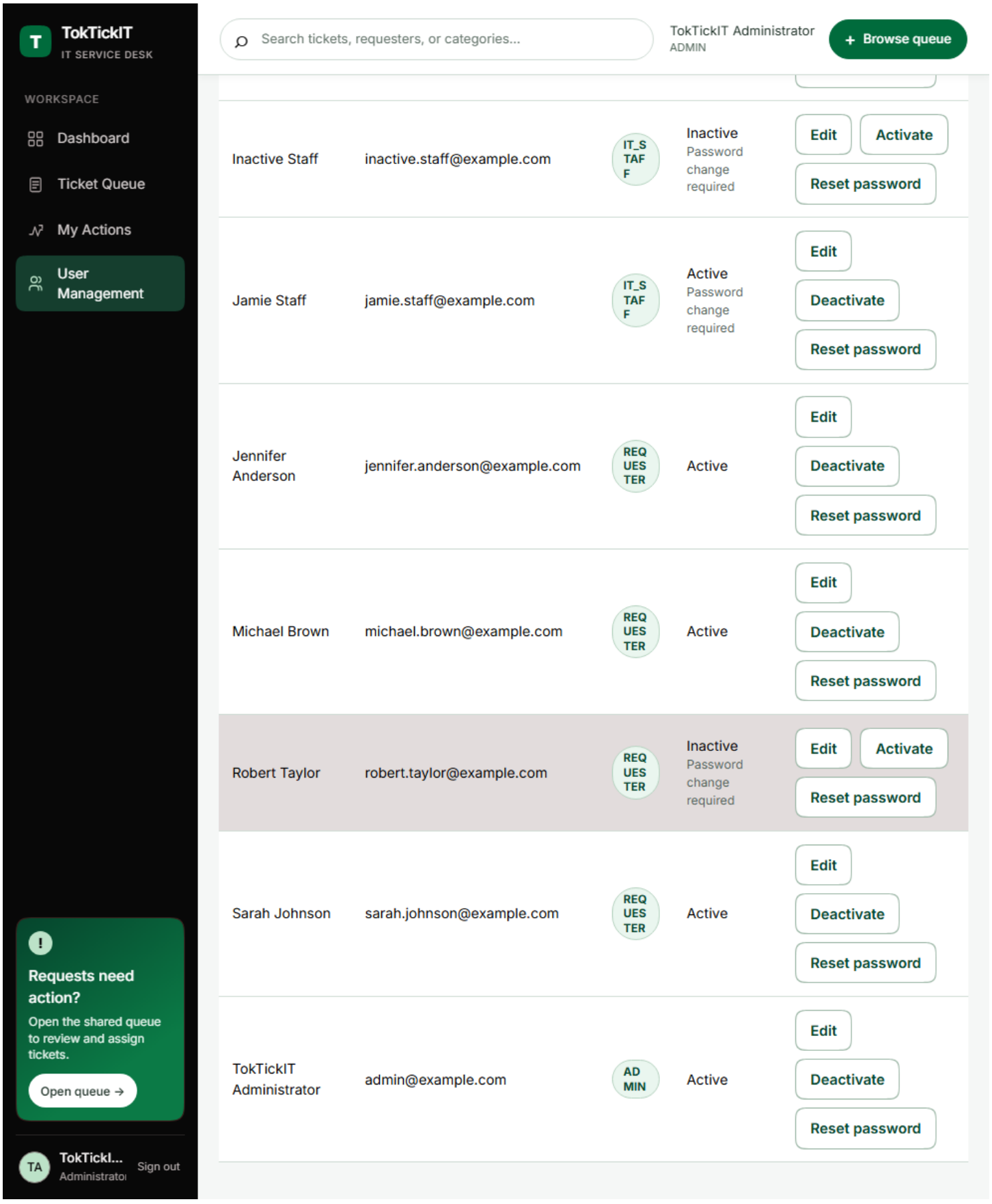
Administrator-only account management remains part of the inherited release.

User Management · Tablet top (991px)



Real 991 × 1200 browser viewport; all five table columns including Actions fit on screen.

User Management · Tablet lower (991px)



Actual lower browser viewport after vertical scrolling; controls remain accessible and the page width stays at 991px.

Answer Part 9: Final Release Summary and Readiness (5 Points)

Lab 4 Issues #51–#56, UX/UI Issue #63 / PR #64, and responsive Issue #65 / PR #66 are closed and merged. PR #66 was formally approved and merged to main at `a76ee09dc558aa2efcd6f4ef3f3bd759ba3bed7e`, now verified as the main branch tip. Client and full E2E results cover the overlap correction. Server tests/build were freshly rerun at source commit `ca82049e1681`; PR #66 contains no server changes, so the tested server source is unchanged at final main. GitHub returned no status checks or workflow runs. The Project board state and the student's own review of the reflection still need confirmation.

9.1 Rendered UI specification and accessibility checklist

Contract	Verified design rule
Shared system	Reuse Lab 2/3 Zen Green tokens, forms, tables, shared shell and focus rings; no second palette.
Typography	Bundled Inter with system fallback; documented 24px/18px/15px/14px/12px hierarchy.
Responsive	Desktop ≥992px; tablet 768–991px; mobile <768px. Reflow components; page itself never overflows horizontally.
Navigation	Sticky desktop/tablet navigation; mobile disclosure is keyboard operable, closes on Escape and restores focus.
Accessibility	Named controls, visible focus, field-linked errors, live mutation feedback, non-color status cues, reduced-motion support.
Visual review	PR #66 adds no-overlap and no-overflow assertions at 768, 991 and 992px; its full E2E record reports 36 passed across desktop/tablet/mobile.

Core Zen Green tokens and type scale

Token	Value / use
Primary / focus green	#006B3C / #0B7A46
Pale / page / surface	#EAF6EF / #F5F7F6 / #FFFFFF
Primary / secondary text	#1B2E24 / #52665A
Neutral border / error	#D1DDD6 / #C53030
Typography	Inter; 24 / 18 / 15 / 14 / 12px heading-to-caption scale.

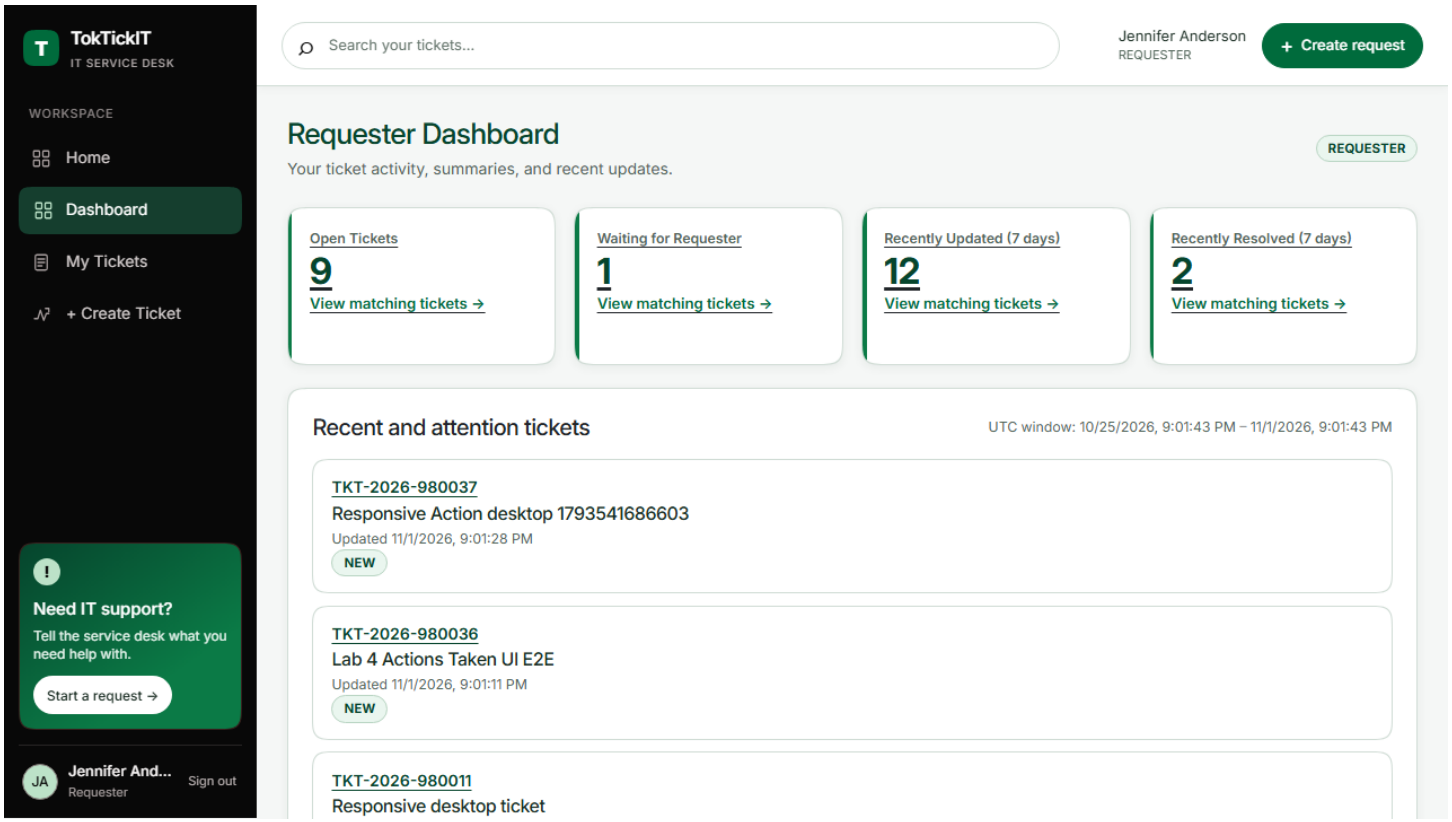
Full committed ui-spec.md

9.2 Lab 4 screen evidence · desktop, tablet and mobile

9.2 Responsive viewport evidence · Requester Dashboard

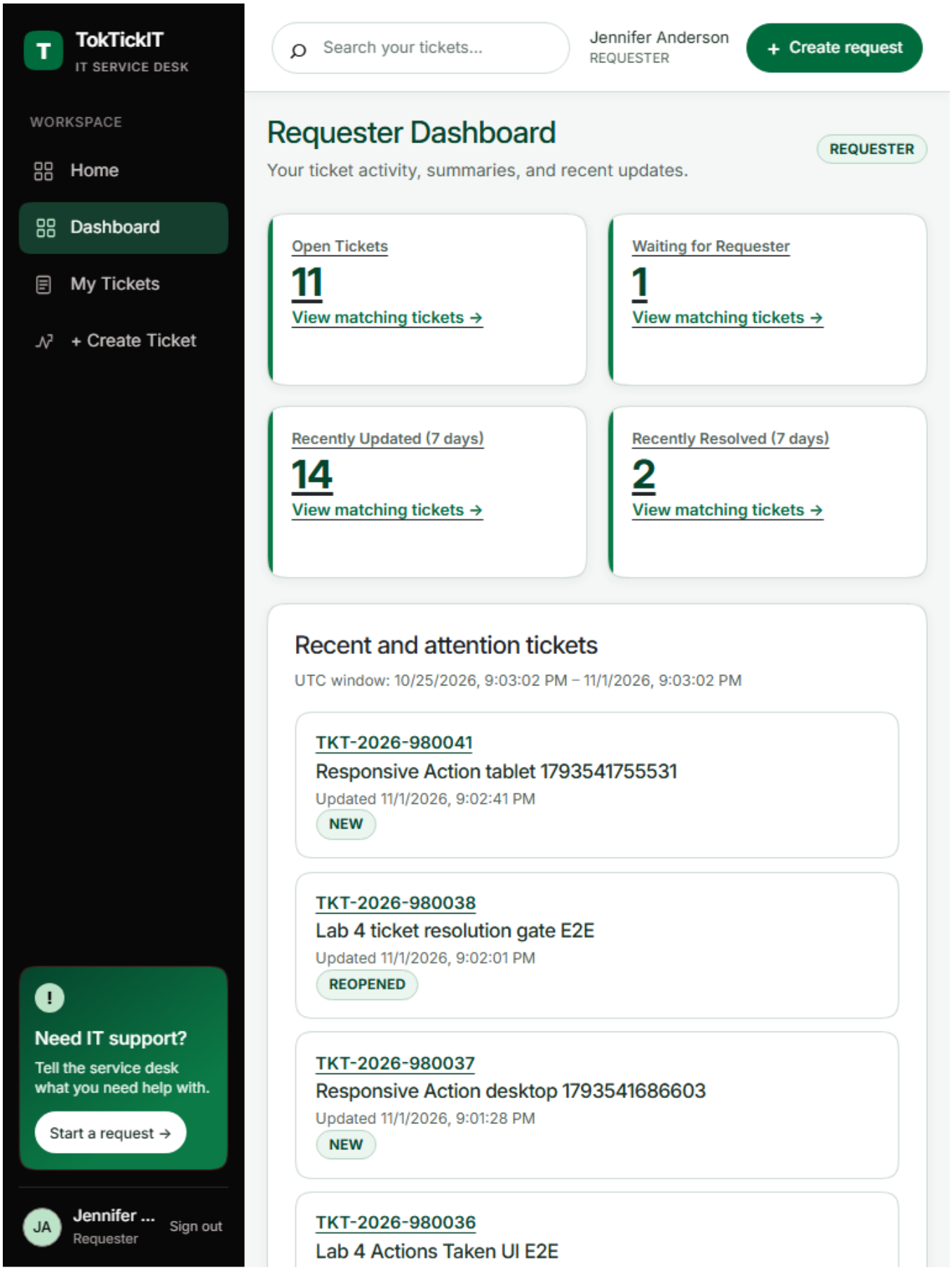
Role-scoped dashboard and ticket list; see the same route at all three required viewport bands.

Requester Dashboard · Desktop



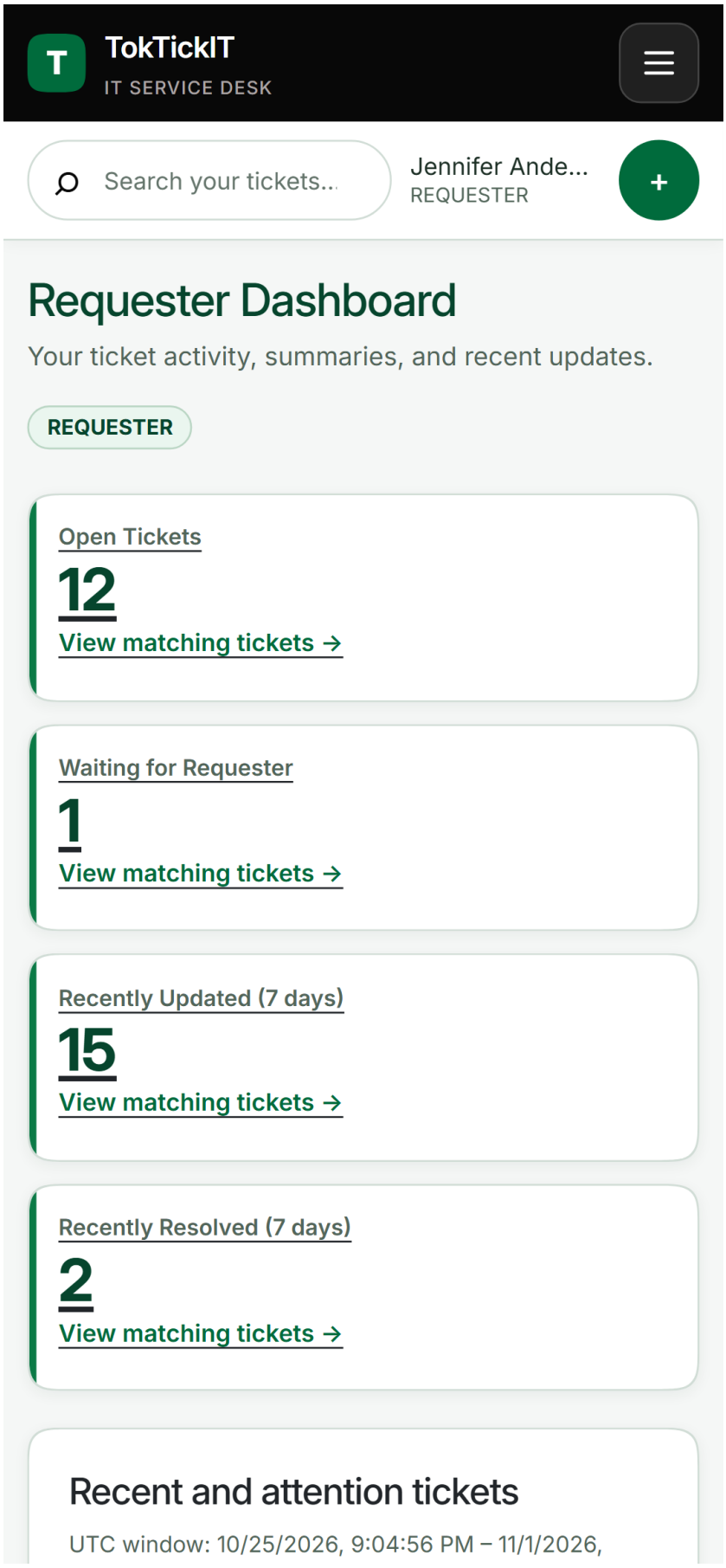
Real Playwright desktop viewport capture; source was recorded by the current full E2E run.

Requester Dashboard · Tablet



Real Playwright tablet viewport capture; source was recorded by the current full E2E run.

Requester Dashboard · Mobile

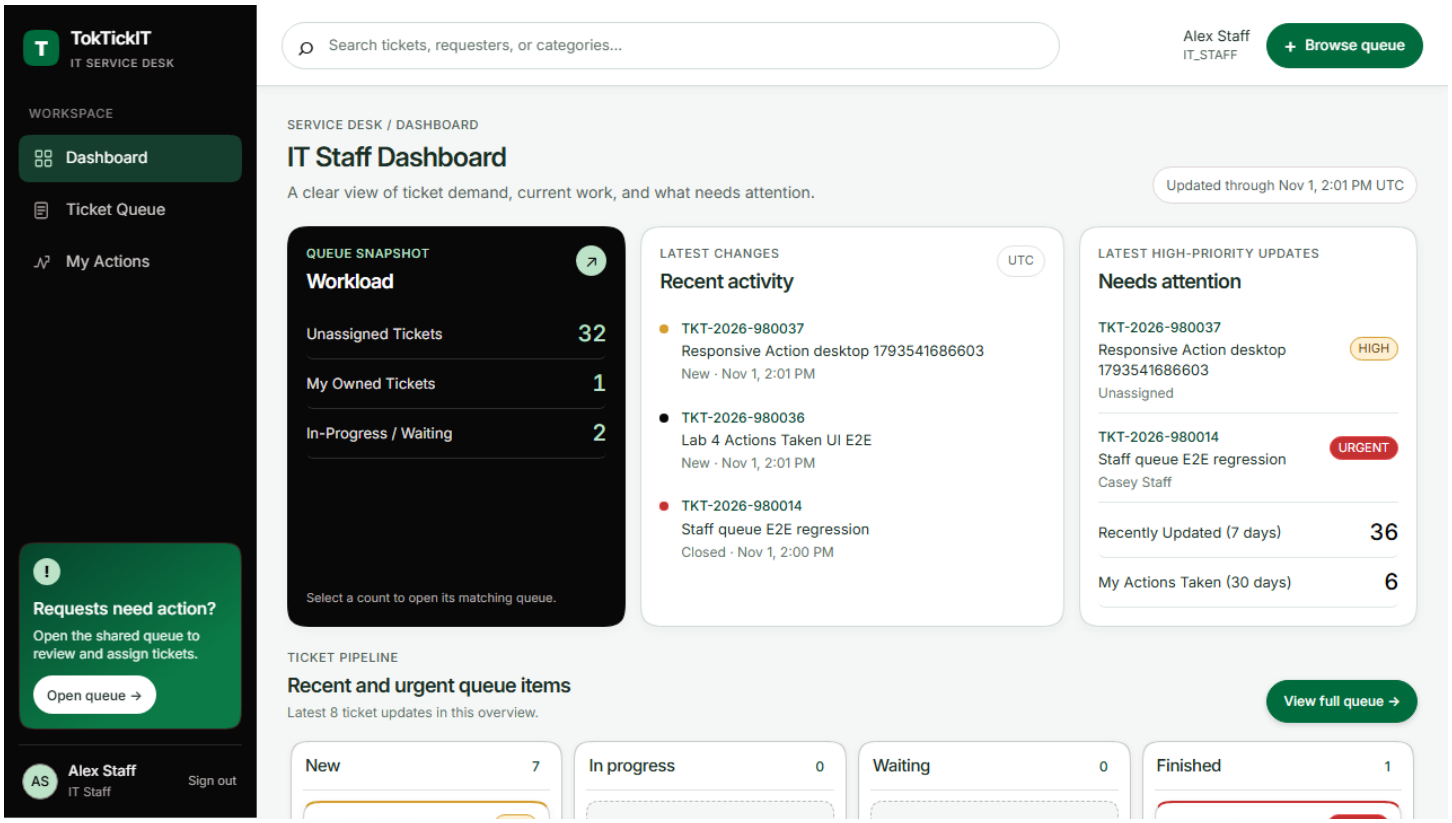


Real Playwright mobile viewport capture; source was recorded by the current full E2E run.

9.2 Responsive viewport evidence · IT Staff Dashboard

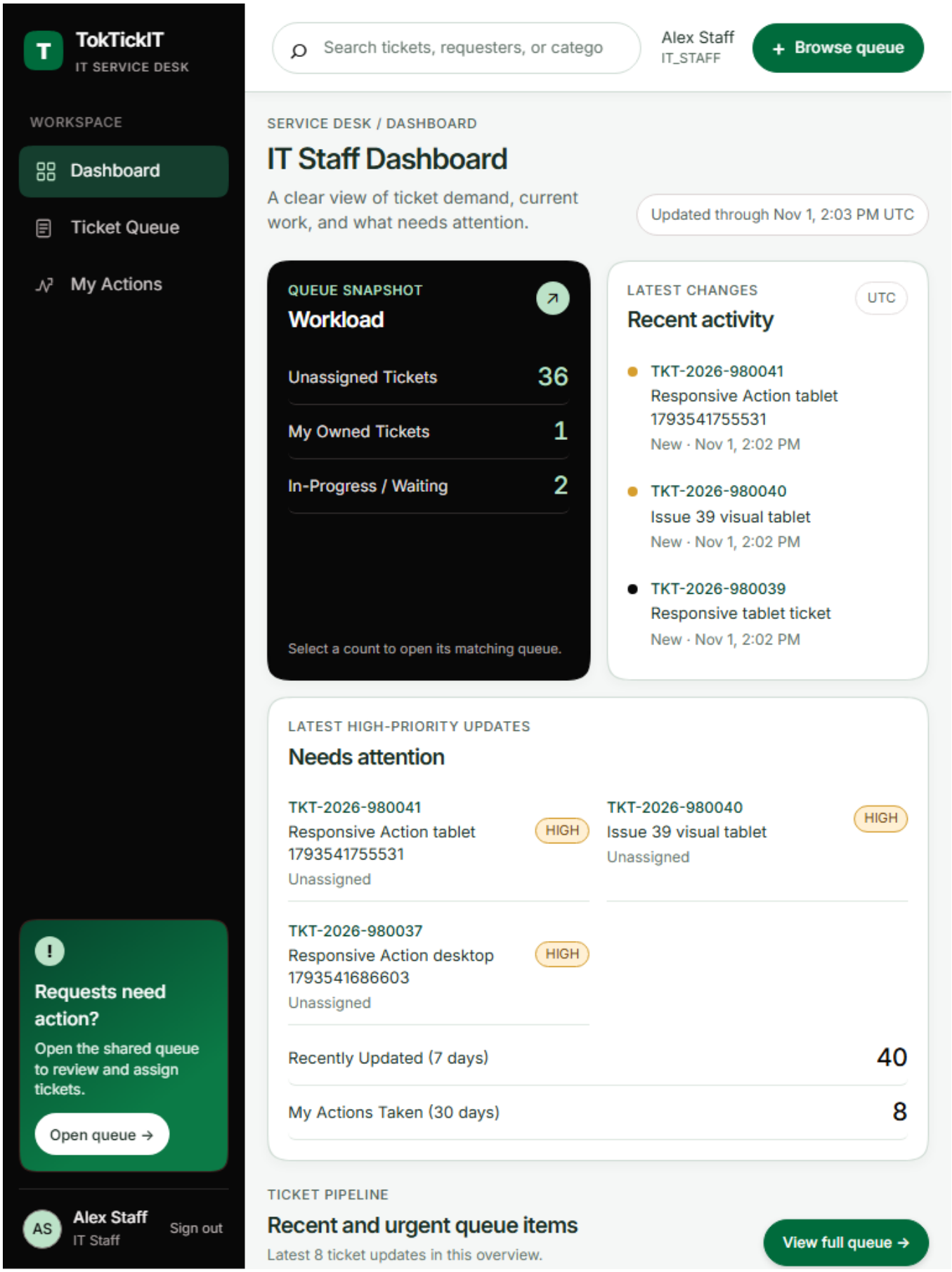
Operational metrics, queue drill-downs and recent/urgent work at desktop, tablet and mobile widths.

IT Staff Dashboard · Desktop



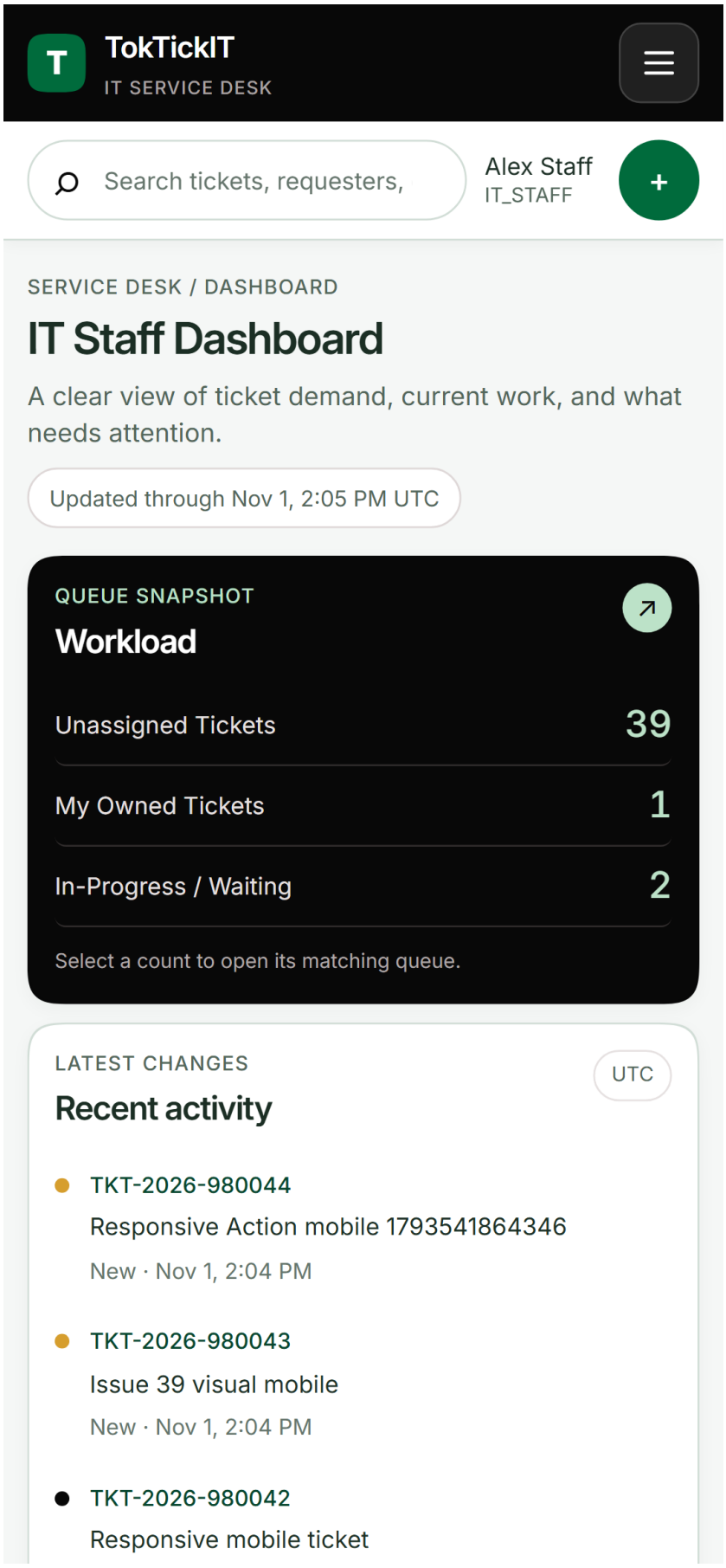
Real Playwright desktop viewport capture; source was recorded by the current full E2E run.

IT Staff Dashboard · Tablet



Real Playwright tablet viewport capture; source was recorded by the current full E2E run.

IT Staff Dashboard · Mobile

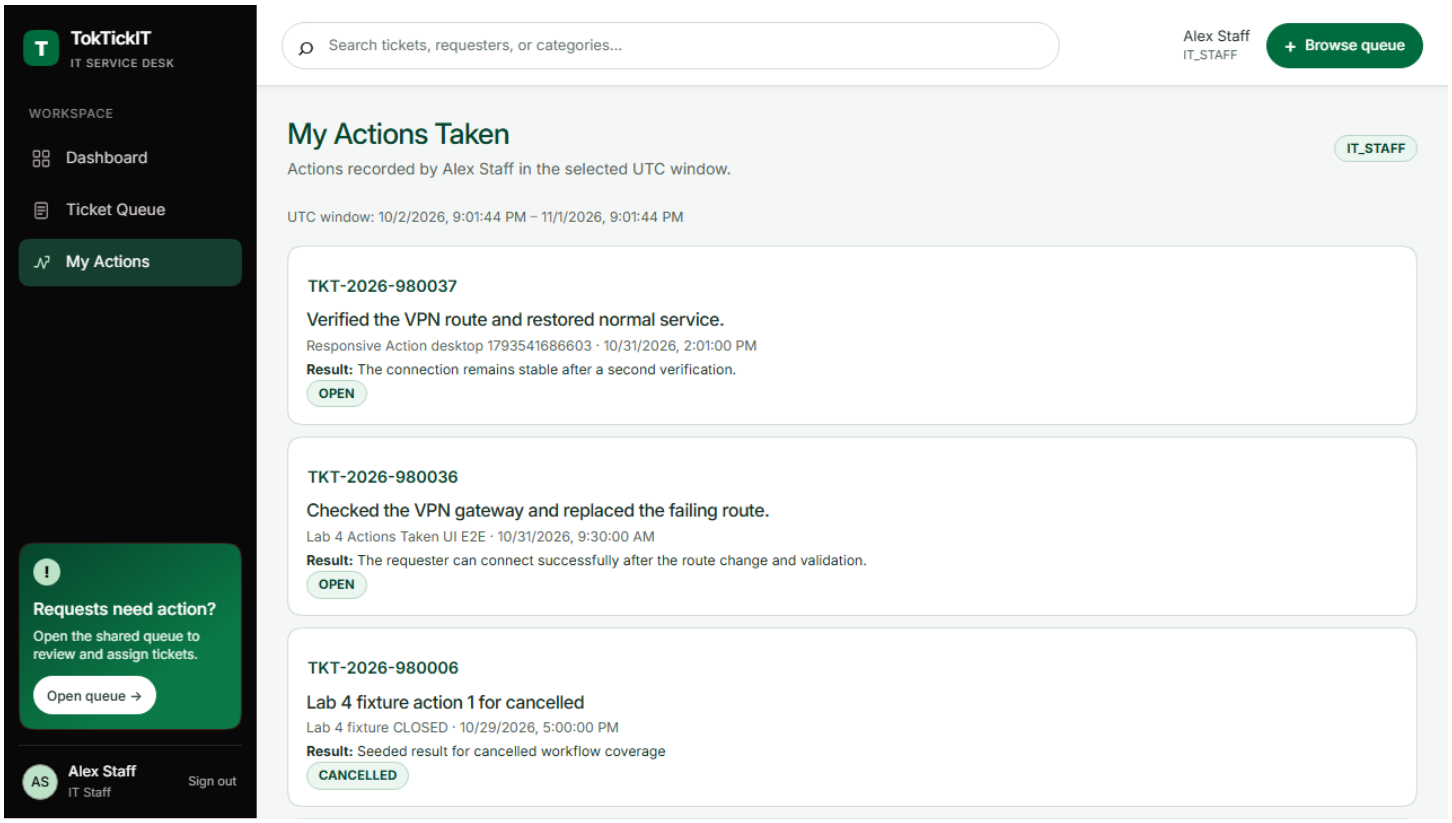


Real Playwright mobile viewport capture; source was recorded by the current full E2E run.

9.2 Responsive viewport evidence · My Actions Taken

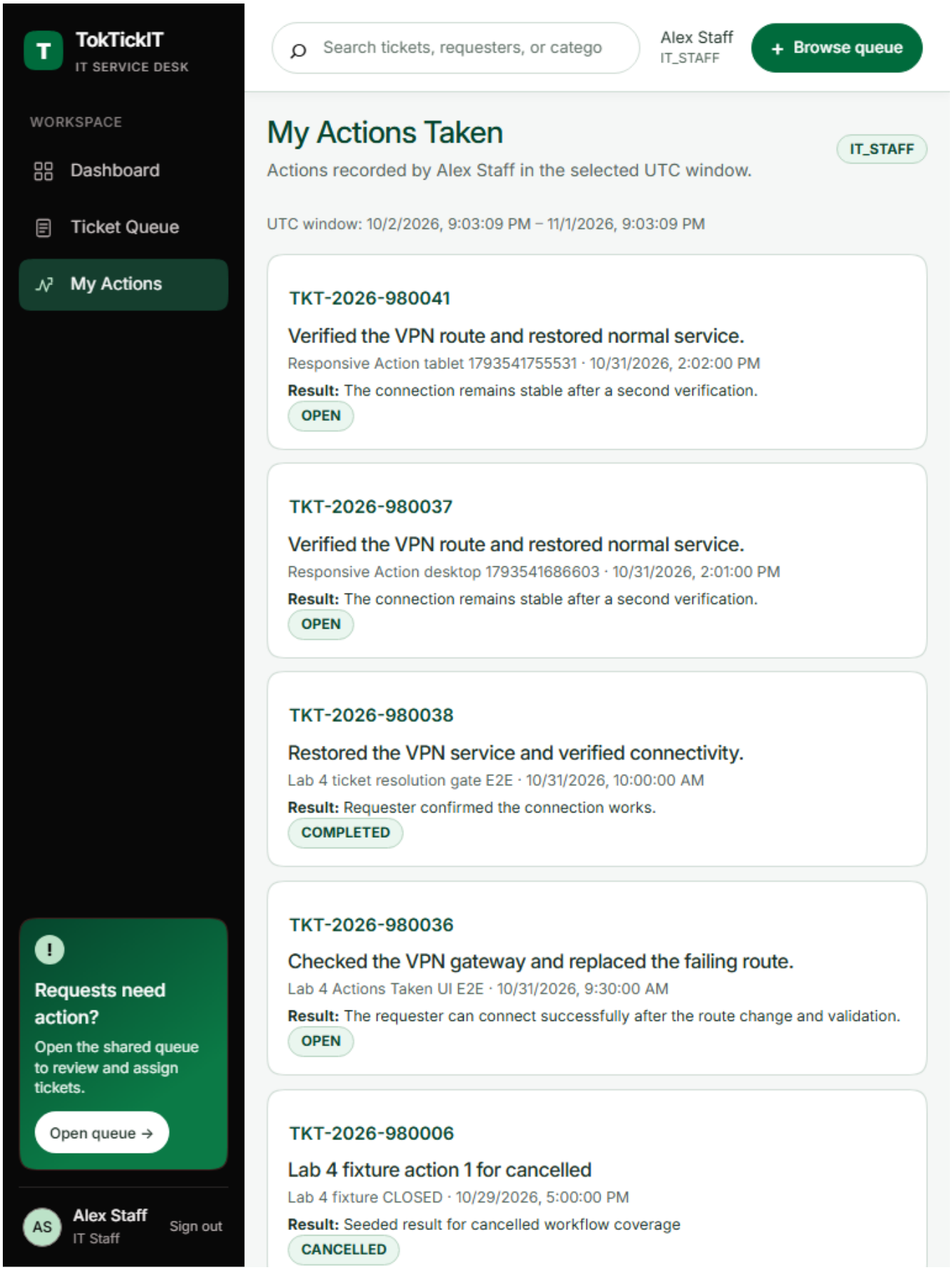
Authenticated staff Action list and date/user-scoped results at each breakpoint.

My Actions Taken · Desktop



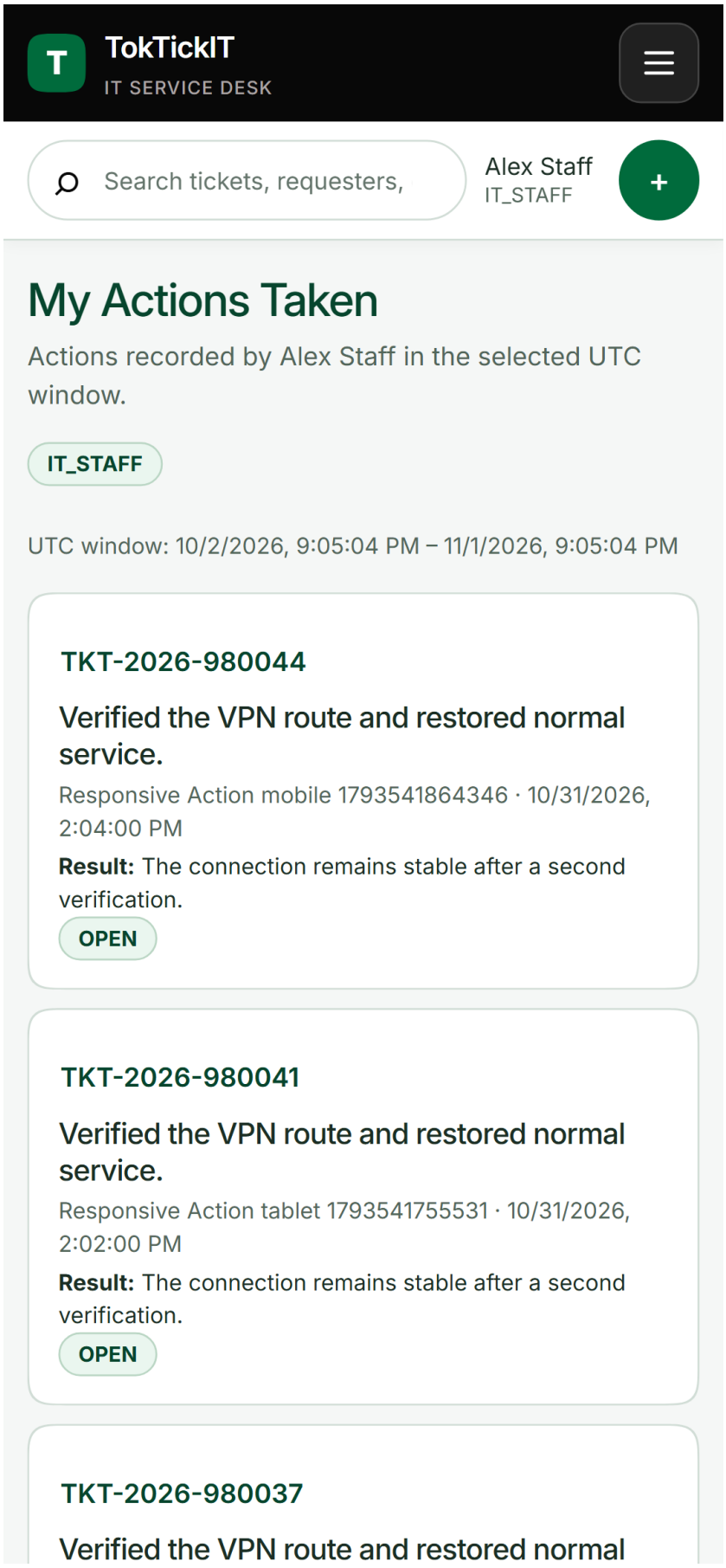
Real Playwright desktop viewport capture; source was recorded by the current full E2E run.

My Actions Taken · Tablet



Real Playwright tablet viewport capture; source was recorded by the current full E2E run.

My Actions Taken · Mobile

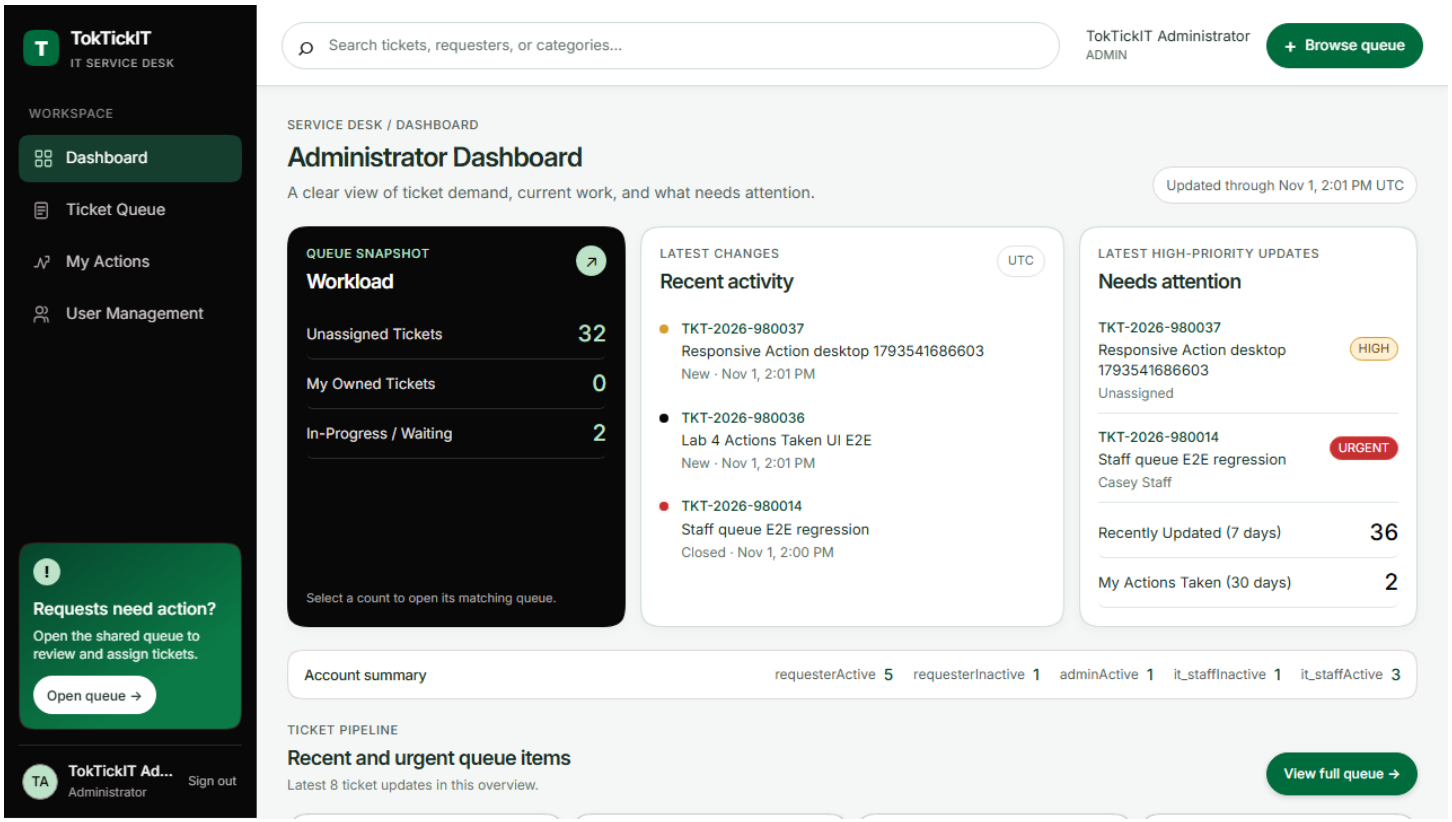


Real Playwright mobile viewport capture; source was recorded by the current full E2E run.

9.2 Responsive viewport evidence · Administrator Dashboard

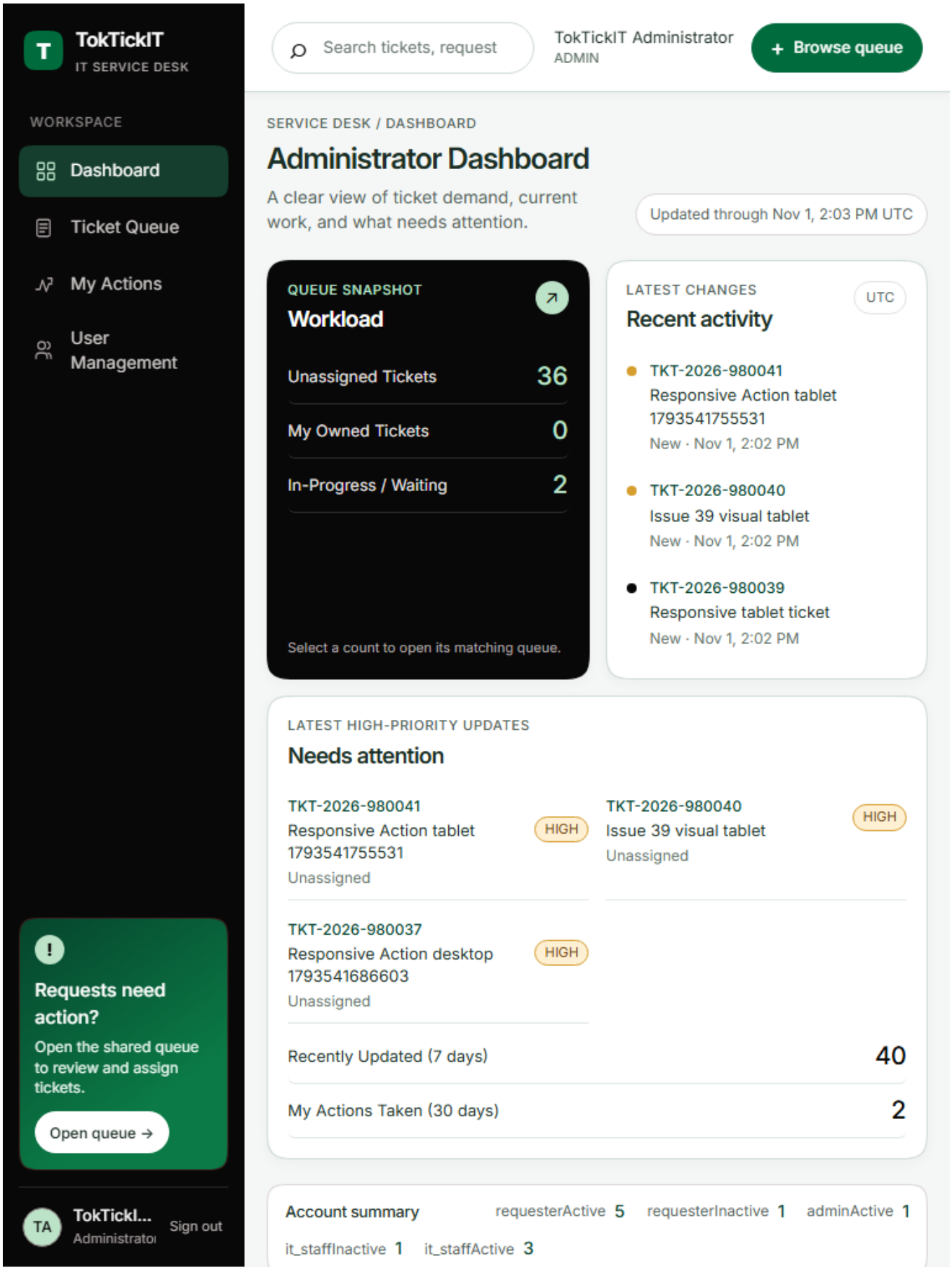
Administrator's authorized operational summary at each breakpoint.

Administrator Dashboard · Desktop



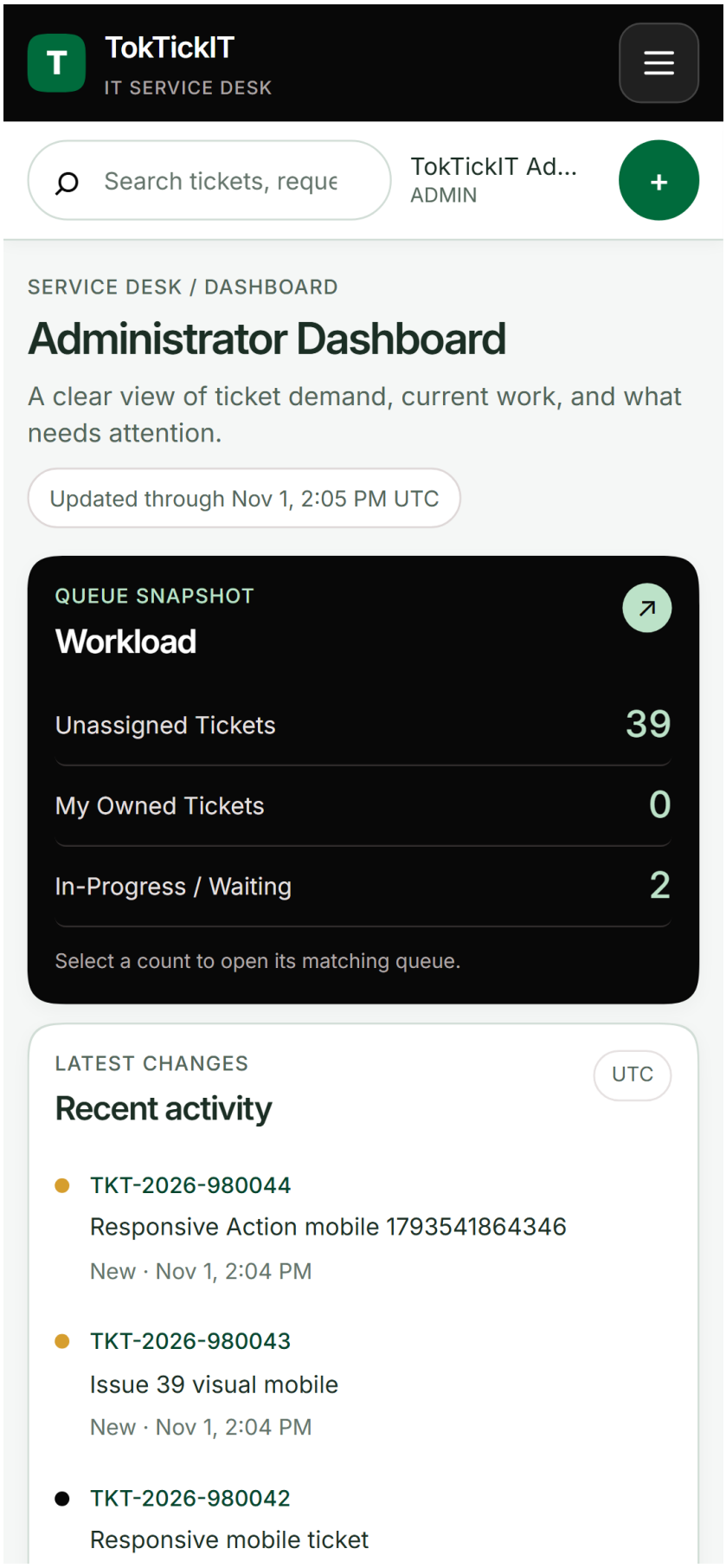
Real Playwright desktop viewport capture; source was recorded by the current full E2E run.

Administrator Dashboard · Tablet



Real Playwright tablet viewport capture; source was recorded by the current full E2E run.

Administrator Dashboard · Mobile

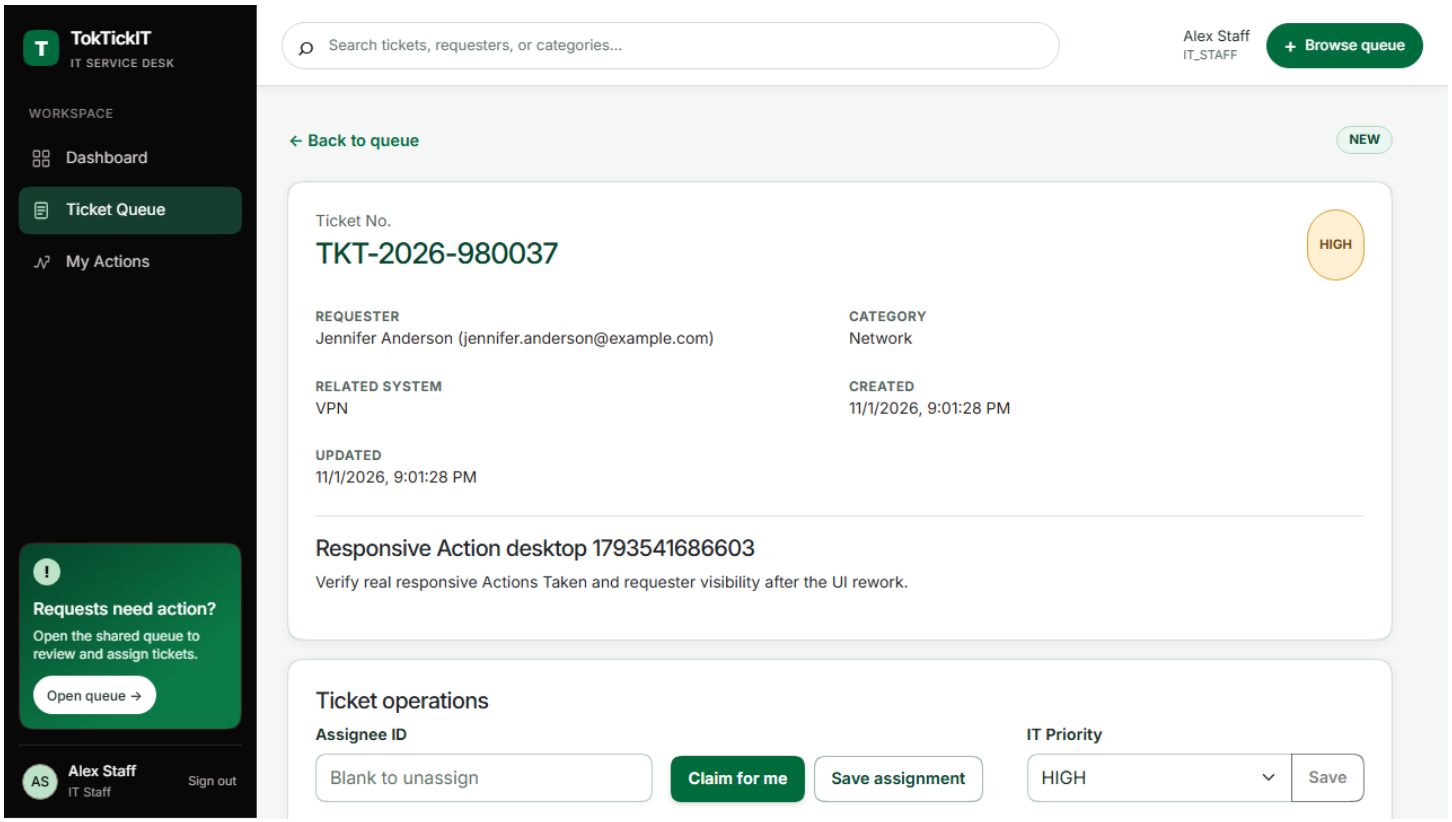


Real Playwright mobile viewport capture; source was recorded by the current full E2E run.

9.2 Responsive viewport evidence · Staff Ticket Detail and resolution gate

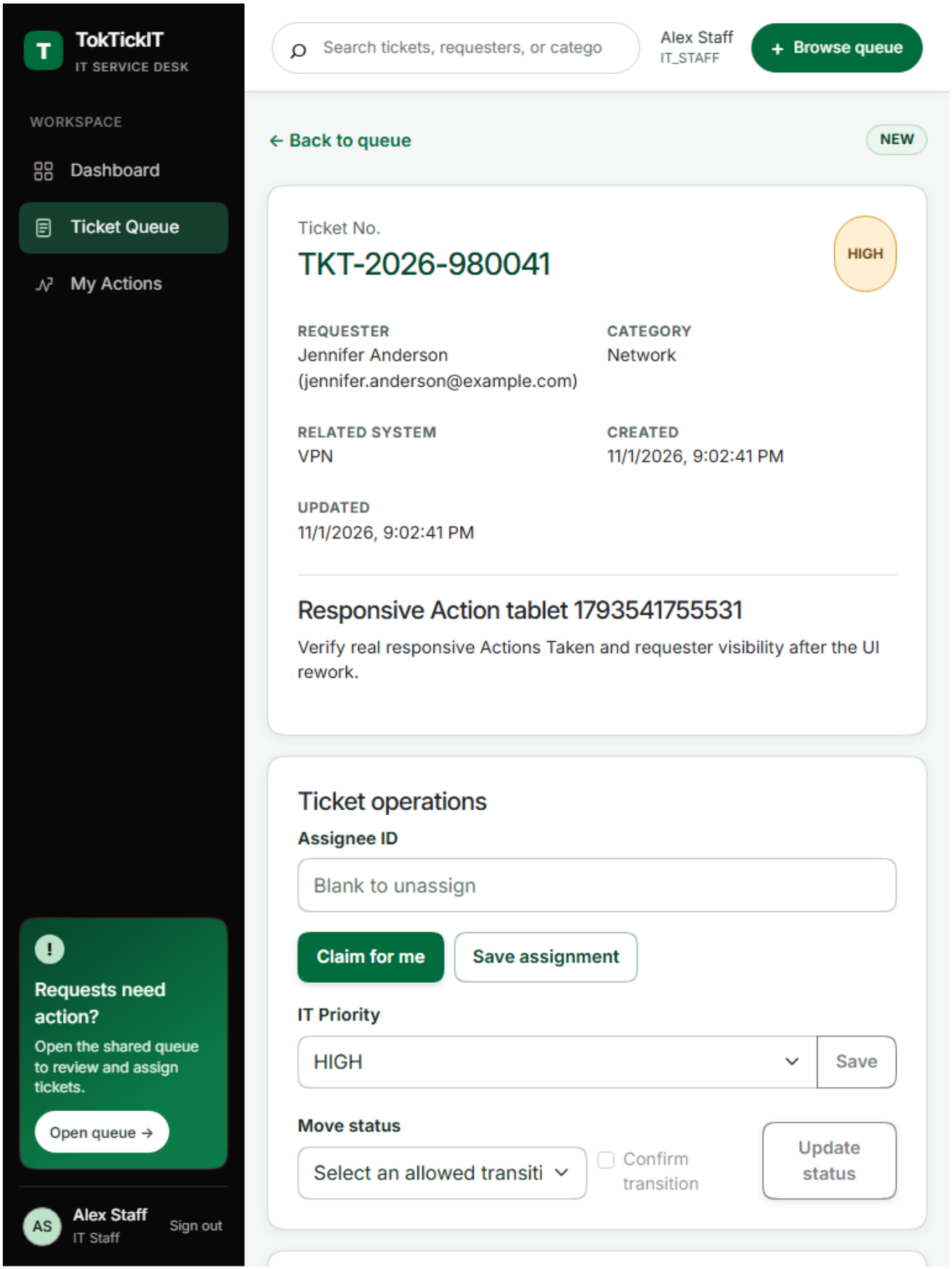
Ticket operations, Actions Taken and requester/staff communication at each breakpoint.

Staff Ticket Detail and resolution gate · Desktop



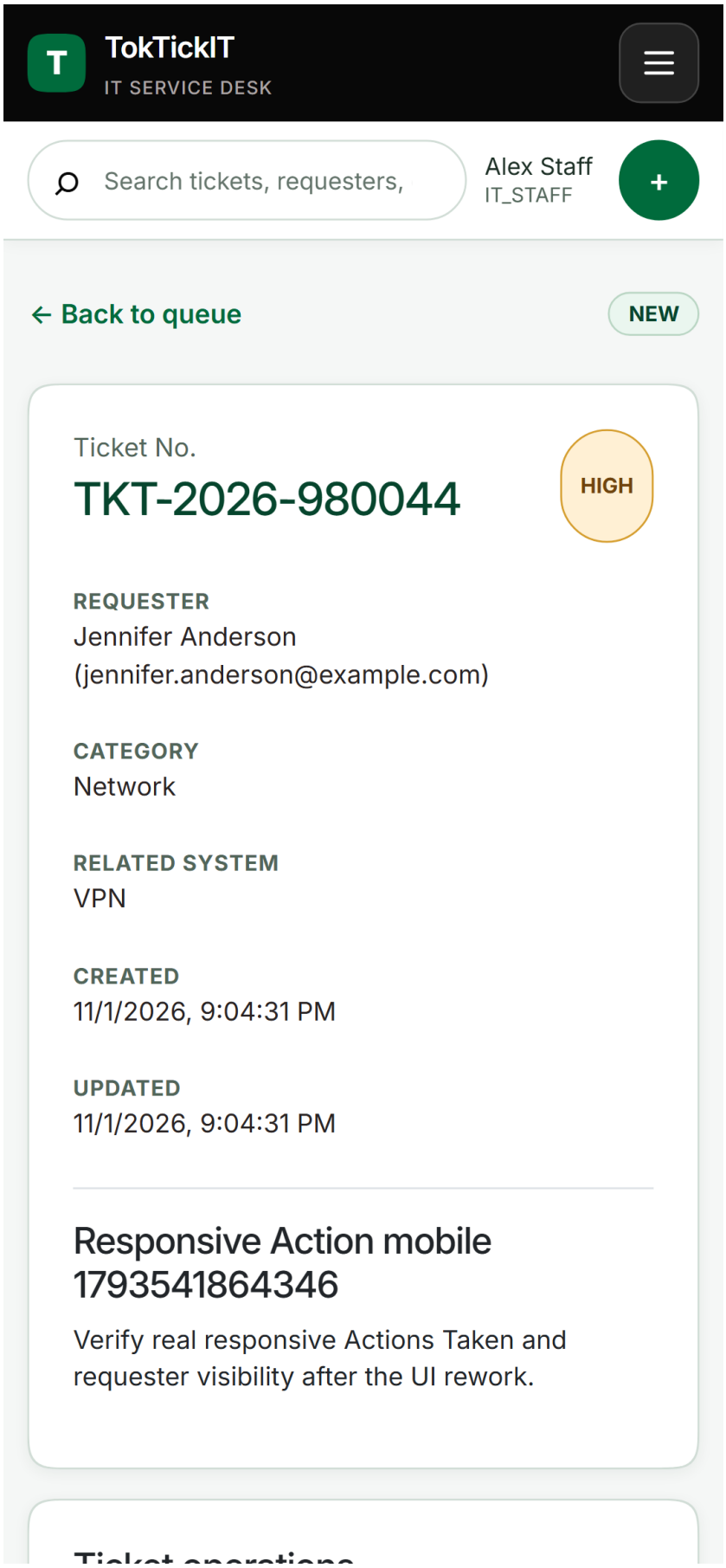
Real Playwright desktop viewport capture; source was recorded by the current full E2E run.

Staff Ticket Detail and resolution gate · Tablet



Real Playwright tablet viewport capture; source was recorded by the current full E2E run.

Staff Ticket Detail and resolution gate · Mobile



Real Playwright mobile viewport capture; source was recorded by the current full E2E run.

9.3 Assignment gate status

Gate	Current evidence	Status
Six ordered Issues / Kanban workflow	Issues #51–#56 and PRs #57–#62 closed/merged; UI follow-up Issues #63/#65 and PRs #64/#66 also closed/merged. Project card columns remain unreadable.	Verify board
Spec DD / Test DD	Numbered requirements, API/UI contracts, AC mapping and named test sources are committed on main.	Evidence included
Core Lab 4 and UI regression	Server 44 files / 136 passed; build passed at ca82049e1681. The PR #66 diff has no server files; client 73 passed and Playwright 36 passed.	Verified source; no CI
UX/UI rework	Issue #63 closed; PR #64 approved and merged at e22a50e3a932.	Merged
Tablet responsive correction	Issue #65 closed; PR #66 approved and merged at a76ee09dc558; this is the current main tip.	Merged
UI responsive/accessibility evidence	128 post-PR #64 captures plus 27 release and 28 compliance captures from the responsive follow-up; E2E spans desktop/tablet/mobile.	Evidence included
GitHub checks	No commit status checks or workflow runs returned for the merge SHA.	No CI result
Student reflection / final PDF	Report regenerated with current merge and test evidence; student should verify the reflection and final wording.	Student review

9.4 Final actions before submission

• Confirm the Issue #63 and #65 cards are in the expected final columns on the Project board; the connected GitHub API could not read Project status.

• Read the AI-use reflection and adjust it in your own words so it accurately describes your work.

• Keep the real browser screenshots and their current manifest with the PDF if the instructor expects the source evidence separately; the report embeds representative Lab 2–4 captures.

• Review all nine parts and links in the refreshed PDF before uploading the canonical Lab 4 report.

> The reviewed UI follow-up is merged at the current main tip. Client and E2E results cover the responsive fix. Server tests/build were rerun at the recorded source commit; the final PR changed no server files, so the tested server source is unchanged at main. The remaining student-owned checks are the Project board column and reflection wording. No GitHub CI result or guaranteed grade is claimed.

Lab 4 specification and test contracts